Enterprise Google Cloud Data Platform Blueprint

VOLUME 07

AI, GenAI, RAG and LLMOps

Enterprise Reference Architecture and Implementation Blueprint
Architecture Baseline 1.0 | 11 July 2026

Classification. Enterprise architecture reference. Adapt to workload, jurisdiction, risk, product edition and current Google Cloud capabilities before implementation.

Document Control

This volume defines the enterprise AI platform that consumes the governed data products, documents, semantic assets and controls established in Volumes 01-06. It is a standalone continuation and does not modify prior volumes. Product terminology is current as of the document date but must be revalidated during implementation.

Field	Value
Document	Volume 07 - AI, GenAI, RAG and LLMOps
Series	Enterprise Google Cloud Data Platform Blueprint
Version	1.0
Status	Detailed reference architecture for adaptation and implementation
Date	11 July 2026
Owner	Enterprise AI Platform Architecture
Approvers	CIO, CDO, Enterprise Architecture, Security, Privacy, Responsible AI, Legal, Data Governance, Platform Engineering, SRE and FinOps
Audience	Executives, AI/ML leaders, architects, data scientists, ML engineers, application and agent developers, security, privacy, risk, audit and operations
Review cycle	Quarterly and after material model, product, law, risk, location, security or commercial change
Primary dependencies	Volumes 01-06; Volume 08 Security; Volume 09 DevSecOps; Volume 10 FinOps; Volume 11 SRE; Volume 12 DR; Volume 13 Terraform
Scope	Enterprise AI strategy, MLOps, GenAI, RAG, agents, evaluation, security, responsible AI, operations and implementation examples

Current product terminology. The master prompt uses Vertex AI Agent Builder. Current Google Cloud documentation groups agent-building capabilities under Gemini Enterprise Agent Platform and identifies Agent Development Kit, Agent Runtime, Agent Search, RAG Engine and related services. This volume preserves the master-prompt concept while using current names and explicitly marking product/location/provider validation as an implementation dependency.

Guidance Classification

Classification	Meaning
Mandatory baseline	Required unless a documented architecture, security and risk exception is approved.
Recommended pattern	Default for most workloads; alternatives require selection evidence.
Conditional pattern	Adopt only when workload criteria, product availability and economics support it.
Future-state capability	Target direction not required for the initial production release.
Open architecture decision	Must be resolved before the affected implementation or release.

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1. Executive Summary

The target state is a governed enterprise AI platform in which predictive models, generative applications, retrieval systems and agents are built from authorized data products and released through evidence-based controls. Google Cloud managed services are used serverless-first, while security, responsible-AI review, evaluation, observability, cost allocation and recovery are treated as product requirements rather than post-production additions.

The platform separates four concerns: the data and knowledge foundation; model and reasoning capabilities; agent/application experiences; and the trust/operations control plane. It supports classic ML with Vertex AI Pipelines, Experiments, ML Metadata, Model Registry, Feature Store and Model Monitoring; generative AI with approved Gemini models, prompt lifecycle and evaluation; enterprise RAG with permission-aware retrieval, citations and evidence lineage; and agentic systems with narrow tools, workload identities, policy gateways and human approval for high-impact actions.

Executive decision	Approved direction	Business outcome
Primary AI platform	Use Gemini Enterprise Agent Platform / Vertex AI services as the managed enterprise AI plane, integrated with inherited BigQuery, Cloud Storage and governance.	One governed path from data to models, RAG and agents.
AI consumption	Consume certified data products and approved document corpora, never unmanaged raw data by default.	Higher trust, lineage and regulatory defensibility.
Model choice	Select models from an approved catalog by quality, safety, location, latency, cost and portability—not prestige alone.	Fit-for-purpose economics and controlled risk.
RAG	Use permission-aware retrieval with source ACLs, metadata filtering, grounding, citations and deletion propagation.	Useful answers without bypassing source authorization.
Agents	Use ADK/Agent Runtime patterns with allowlisted tools and confirmation for consequential actions.	Automation with bounded autonomy and auditability.
Release governance	Evaluation, security testing and risk-tier approvals are mandatory release gates.	Reduced unsafe, ungrounded or noncompliant production behavior.

Executive acceptance condition. No production AI use case is approved solely because a model demo performs well. Approval requires a named business owner, risk tier, authorized data, measurable benefit, evaluation evidence, security design, operating SLO, unit economics, human-oversight model and tested rollback.

2. Inherited Context, Scope and Assumptions

This volume inherits the Fortune 100 scale, hybrid/multi-cloud estate, regulated jurisdictions, security classifications, AMER/EMEA/APAC regional strategy, project factory, Shared VPC, group-based IAM, VPC Service Controls, selective CMEK, centralized logging, Terraform authority and federated data-product model established in the preceding volumes.

Inherited element	Volume 07 implication
Certified data products	Training, retrieval, feature and evaluation inputs reference product IDs, owners, contracts, quality scores and residency.
Regional data planes	Models, corpora, endpoints, pipelines and telemetry are regionalized unless an explicit cross-region decision is approved.
Data classifications	Public, Internal, Confidential, Restricted and Regulated drive model eligibility, retrieval filters, retention, human review and logging.
Identity model	Human access uses enterprise groups; runtime access uses dedicated workload identities with no embedded credentials.
Network model	Private access, controlled egress and service perimeters apply to sensitive AI workloads; capabilities are validated service by service.
Analytics semantic layer	Agents and copilots may call certified Looker/BigQuery semantic interfaces rather than recreating enterprise KPI logic.
Data mesh	Domains own AI products and meaning; the AI platform team owns reusable capabilities and guardrails.
SLO tiers	Tier 0-2 RTO/RPO and availability objectives are mapped to each AI service and use case.

2.1 Confirmed scope

- Predictive ML, batch/online inference and MLOps.
- Gemini integration, prompt lifecycle, evaluation and LLMOps.
- Permission-aware RAG, enterprise search, embeddings and vector retrieval.
- Single-agent and multi-agent architectures, tools, memory policy and human approval.
- Document AI, speech, vision and multimodal use cases.
- Security, privacy, responsible AI, audit, SLOs, FinOps and DR.

2.2 Architecture assumptions

ID	Assumption	Validation trigger
AI-A01	Enterprise identity, project factory, CI/CD, logging and KMS foundations are available.	Before first nonproduction deployment.
AI-A02	Priority domains can supply certified, contract-bound training/retrieval inputs.	Use-case intake.
AI-A03	No single model or vector backend satisfies every workload.	Each architecture review.
AI-A04	Model and platform product names, quotas, regions and controls can change.	Quarterly and before production.
AI-A05	High-impact decisions retain meaningful human oversight unless explicitly authorized.	Risk-tier review.
AI-A06	AI telemetry is itself sensitive and subject to minimization, access and retention.	Logging design.
AI-A07	The enterprise will maintain coexistence with selected third-party/open-source models where justified.	Model catalog decision.

3. Enterprise AI Strategy

The strategy is value-led and control-led: start with measurable business workflows, use the smallest sufficient model and architecture, ground decisions in governed enterprise context, automate evaluation and deployment, and continuously measure quality, safety, adoption and unit economics. AI is managed as a portfolio of products, not a collection of experiments.

Strategic pillar	Concrete architecture decision	Success measure
Business value	Every use case has accountable benefit metrics and termination criteria.	Realized benefit and adoption versus approved case.
Governed context	Models and agents consume certified products/corpora with lineage.	Percentage of production inputs with owner, contract and quality evidence.
Managed-first	Prefer managed platform capabilities before self-managed serving/vector infrastructure.	Lower operational toil and faster recovery.
Risk proportionality	Controls and evidence scale by impact, autonomy, audience and data class.	100% risk classification and approval completeness.
Evaluation-first	Golden/adversarial datasets and release thresholds precede production.	Release gate pass rate and escaped-defect rate.
Reversible delivery	Version model, prompt, tools, indexes and configuration independently.	Rollback success and recovery time.
Economics	Track unit cost per successful business task, not just token or endpoint spend.	Cost variance and value/cost ratio.

3.1 Portfolio categories

Category	Representative workloads	Default pattern
Predictive ML	Churn, demand, fraud, propensity, forecasting	BigQuery/Vertex AI training, Registry, batch or private endpoint, monitoring.
Generative assistance	Drafting, summarization, extraction, coding assistance	Approved Gemini model, prompt contract, safety and evaluation.
Enterprise RAG/search	Policy, product, case and research assistants	Permission-aware corpus, hybrid retrieval, grounding and citations.
Agentic workflow	Case handling, service operations, analytics and approvals	ADK/Agent Runtime, narrow tools, policy gateway and human confirmation.
Multimodal intelligence	Document, voice, image and video workflows	Specialized API/model plus structured extraction and review.

4. AI Use Cases, Personas and Risk Tiering

Persona	Primary responsibility	Platform capability	Control emphasis
Business owner	Own purpose, benefit, affected users and consequences	Use-case registry and scorecard	Accountability, human oversight and termination.
Data scientist	Experiment, train and evaluate	Workbench, Pipelines, Experiments and Registry	Reproducibility and authorized data.
ML engineer	Operationalize training and inference	Pipeline templates, endpoints, monitoring	Reliability, isolation and rollback.
GenAI engineer	Design prompts, RAG and evaluation	Prompt repository, RAG/search and evaluation	Grounding, safety and versioning.
Agent developer	Build orchestration and tools	ADK, Agent Runtime and tool gateway	Least privilege, trajectories and action controls.
Independent validator	Challenge quality, bias, safety and resilience	Read-only evidence, test sets and reports	Independence and evidence integrity.
AI platform/SRE	Operate shared capabilities	Projects, network, observability and capacity	SLOs, incidents and FinOps.
Security/privacy/legal	Approve risk treatment	Policy, DLP, audit and review workflow	Data protection, misuse and compliance.

4.1 Risk-tier decision factors

- Decision impact and whether an output may affect rights, access, employment, healthcare, finance, safety or regulatory reporting.
- Degree of autonomy, especially whether tools can write, submit, communicate, purchase, delete, approve or change entitlements.
- Audience and scale: internal expert, general workforce, customers, partners or public.
- Data sensitivity, geography, source authorization and possibility of memorization or leakage.
- Model opacity, novelty, external dependency and ability to independently verify results.
- Availability, continuity and error consequences.

Tier	Release baseline	Examples
1 - Low	Owner, approved model, privacy/data check, quality/safety smoke tests, logging and rollback.	Internal low-impact drafting with no sensitive data.
2 - Moderate	Tier 1 plus evaluation suite, model/prompt card, grounding where factual, monitoring and periodic review.	Enterprise search, recommendations, predictive prioritization.
3 - High	Independent validation, red team, fairness/impact review, strict access, human confirmation, enhanced evidence and incident exercise.	Regulated decision support or agent with write tools.
4 - Prohibited/exception	Blocked by policy unless formally approved by enterprise risk/legal authority.	Unlawful discrimination, covert manipulation or uncontrolled autonomous high-impact action.

5. AI-ready Data Platform

AI readiness is achieved by making enterprise data and knowledge discoverable, authorized, versioned, quality-scored and machine-consumable. A lake or warehouse containing large volumes of data is not AI-ready unless training, retrieval and evaluation inputs have stable contracts, provenance, residency, retention and deletion behavior.

Asset class	Required controls	AI use
Structured data products	Owner, contract, quality, lineage, point-in-time semantics, classification and regional location.	Training, batch/online features, analytics tools and evaluation.
Document corpora	Source-of-record ID, version, ACL, effective dates, retention/legal hold, deletion and extraction quality.	Search, RAG and agent knowledge.
Feature products	Entity keys, timestamp semantics, transformation code, skew tests and online/offline consistency.	Low-latency prediction and reusable feature engineering.
Evaluation datasets	Representative and adversarial cases, expected behavior, protected access and immutable versions.	Release gates, regression and red-team testing.
Prompts/tools/config	Owner, version, schema, model alias, safety rules and approval.	Reproducible GenAI and agent behavior.
Embeddings/indexes	Source version, model/version, chunking parameters, ACL metadata and deletion propagation.	Retrieval and similarity.

5.1 Point-in-time correctness

Training and evaluation must reconstruct the information available at the prediction decision time. Feature leakage, post-outcome labels and inconsistent entity timestamps invalidate offline performance. Pipelines therefore preserve event time, extraction cutoff, feature transformation version and source-product snapshot identifiers.

5.2 Data minimization and deletion

Only attributes required for the approved purpose are used. Privacy deletion and source ACL changes propagate to training eligibility, corpora, indexes, caches, traces and derived artifacts according to legal requirements and technical feasibility. Immutable compliance evidence is separated from retrievable user content.

6. Target Enterprise AI Architecture

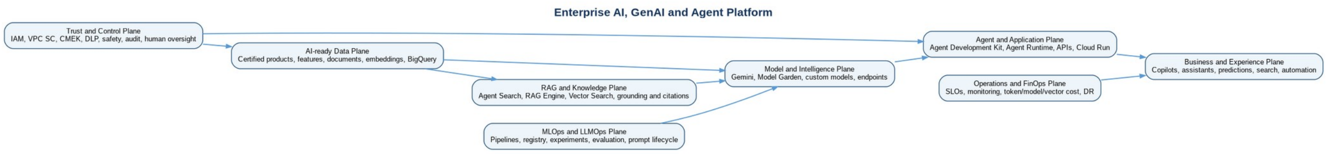

Figure 6-1. Enterprise AI, GenAI and agent platform with explicit data, intelligence, agent, trust and operations planes.

Plane	Core capabilities	Ownership
Experience	Copilots, search, predictions, embedded APIs and workflow automation.	AI product/domain teams.
Agent/application	ADK orchestration, Agent Runtime or application runtime, API policy and tools.	AI product + platform.
Model/intelligence	Gemini, Model Garden, custom/AutoML models, endpoints and batch prediction.	AI platform + ML teams.
Knowledge/RAG	Agent Search, RAG Engine, Vector Search, corpora, grounding and citations.	Knowledge/domain owners + AI platform.
MLOps/LLMOps	Pipelines, experiments, registry, metadata, prompt/config registry, evaluation and promotion.	AI platform/DevSecOps.
AI-ready data	BigQuery, Cloud Storage, features, documents and governed data products.	Data platform/domain owners.
Trust/control	IAM, VPC SC, CMEK, DLP, safety, approval and audit.	Security/privacy/responsible AI.
Operations/FinOps	Monitoring, SLOs, incidents, capacity, token/vector/model unit cost and DR.	SRE/FinOps/product.

6.1 Project and environment pattern

Project pattern	Purpose	Key isolation
acme-ai-platform-<env>-<region>-01	Shared pipelines, registry policies, approved templates, evaluation and platform telemetry.	Platform admin separate from domain data access.
acme-ai-<domain>-<env>-<region>-01	Domain model/RAG/agent development and runtime.	Domain, environment and region boundary.
acme-ai-eval-<env>-<region>-01	Independent evaluation datasets, judge configs and evidence.	Validator write access separated from builders.
acme-ai-security-prod-global-01	Policy, audit aggregation, red-team evidence and emergency controls.	Restricted privileged access.

6.2 Logical network flow

User or application traffic enters through an approved identity-aware application/API path. Runtime identities call only approved models, retrieval services and tool gateways. Restricted data does not traverse public endpoints by design; private access and service perimeters are applied where supported and validated. External model or web grounding is disabled for restricted use cases unless explicitly approved, filtered and logged.

7. Gemini Integration and Model Garden

Gemini models are consumed through approved aliases rather than hard-coded model identifiers in application code. The alias maps to a tested model/version, region, generation configuration, safety profile, context policy and cost class. Model changes are treated as releases because the same prompt and data can produce materially different behavior.

Selection criterion	Evaluation question	Evidence
Task quality	Does the model meet domain-specific task and reasoning rubrics?	Golden-set results and human calibration.
Grounding	Does it follow evidence and abstain appropriately?	Groundedness, citation and abstention suite.
Safety/security	Does it withstand misuse, injection and sensitive-data tests?	Safety and red-team results.
Location/compliance	Can the model and related services operate in the required location/control boundary?	Signed service/location assessment.
Latency/throughput	Does it meet P50/P95 and concurrency targets?	Load test using realistic context/tool calls.
Economics	What is cost per successful business task?	Token, cache, retrieval, tool and retry model.
Portability	Can prompts/tools/evaluation be adapted if the model changes?	Abstraction and fallback design.

7.1 Integration standards

- Use central approved-model aliases and pinned SDK/container dependencies.
- Set explicit generation parameters; do not depend on console defaults.
- Separate system policy, developer instruction, user input, retrieved evidence and tool output.
- Use structured output schemas for downstream automation.
- Record model alias, resolved version, prompt version, retrieval/index version and tool set in each trace.
- Use model context caching only after validating privacy, invalidation and economics.

7.2 Alternatives

Third-party and open models may be used when quality, sovereignty, cost, portability or specialized capabilities justify them. They must pass the same risk, evaluation, security, deployment and operations gates. Self-hosting is not the default because it transfers scaling, patching, serving, safety and DR responsibilities to the enterprise.

8. MLOps Architecture

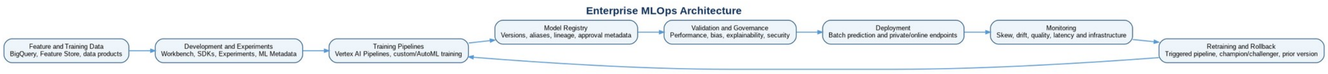

Figure 8-1. Reproducible MLOps lifecycle from governed features through training, registry, validation, deployment, monitoring and rollback.

Lifecycle stage	Mandatory artifact	Control
Experiment	Code commit, dataset snapshot, parameters, environment and metrics.	Experiment/metadata lineage.
Train	Immutable container, pipeline template, runtime identity and output artifact.	CI security and reproducibility.
Evaluate	Quality, segment, fairness, robustness, security and cost results.	Risk-tier thresholds.
Register	Model version, aliases, lineage, card, owner and approval state.	No mutable overwrite.
Deploy	Endpoint/batch config, traffic split, autoscaling, network and CMEK settings.	Separation of duties and canary.
Monitor	Feature/prediction distributions, quality, latency, errors and cost.	Alert and incident workflow.
Retrain/retire	Trigger, new evidence, approval, rollback and archive.	Champion/challenger and lifecycle.

8.1 MLOps versus LLMOps

Dimension	MLOps emphasis	LLMOps extension
Primary artifact	Model weights/container and features.	Model alias plus prompt, corpus/index, tools, policies and generation config.
Evaluation	Predictive metrics and segment/fairness tests.	Task rubrics, grounding, citations, safety, trajectory, latency and cost.
Drift	Feature/prediction/label drift.	Query, corpus, retrieval, prompt, model and behavior change.
Deployment	Batch/endpoint version and traffic.	Coordinated prompt/model/index/agent/tool release.
Observability	Prediction and infrastructure.	Full trace including retrieval and tool calls with privacy controls.

9. Feature Engineering and Vertex AI Feature Store

Feature Store is conditional, not universal. Adopt it when multiple online models require reusable, governed, low-latency features and when point-in-time correctness, feature discovery and online/offline consistency justify another serving layer. BigQuery remains the primary offline feature engineering and training-data plane.

Use Feature Store when	Prefer direct BigQuery/batch features when
Sub-second or low-latency online lookup is required.	Predictions are batch or tolerate query/application latency.
Features are reused across several models and teams.	Features are use-case specific with little reuse.
Online/offline consistency and entity/timestamp semantics are material.	Simple SQL snapshot adequately reconstructs training data.
Operational ownership and monitoring are funded.	Additional serving state would add unjustified complexity.

9.1 Feature standards

- Feature definitions are code, reviewed and linked to source-product contracts.
- Entity keys exclude direct personal identifiers where a surrogate/tokenized key suffices.
- Every feature includes event time, availability time, freshness, null/default policy and allowed use.
- Training joins use point-in-time logic and leakage tests.
- Online values are reconciled with offline source for a sampled entity set.
- Feature drift and missingness are monitored by materiality and segment.

10. Training, Model Registry and Deployment

Capability	Enterprise pattern	Key control
Custom training	Containerized jobs submitted by pipelines with dedicated service accounts.	Pinned dependencies, private data path and resource limits.
AutoML	Conditional accelerator for supported tabular/vision/text workloads.	Same data, evaluation and approval controls as custom models.
Model Registry	System of record for model versions, aliases and deployment lineage.	Immutable version; promotion alias changed only after approval.
Online endpoint	Private/controlled endpoint with autoscaling and traffic split.	Load test, authentication, quotas and fallback.
Batch prediction	Scheduled/event-driven job writing governed result product.	Idempotency, reconciliation and deadline SLO.
Portable serving	Approved container/model artifact for specialized or exit requirements.	Patch, scan, capacity and DR ownership explicit.

10.1 Registry metadata baseline

Metadata	Requirement
Identity	Global use-case ID, model name, version and aliases.
Lineage	Training/evaluation data snapshot, code, pipeline, parameters and environment.
Evaluation	Quality, robustness, bias, security, latency and cost results with thresholds.
Governance	Owner, risk tier, intended/prohibited use, approvals and review date.
Deployment	Endpoint/batch references, traffic, region and runtime dependencies.
Operations	SLO, monitoring, on-call, rollback and retirement state.

11. Batch and Online Prediction

Pattern	Selection criteria	Reliability pattern	Cost pattern
Batch prediction	High-volume scoring, no interactive response, scheduled or event-driven outputs.	Idempotent run ID, input/output reconciliation, partitioned results, deadline alert and rerun.	Right-size accelerators, use regional co-location and shut down ephemeral resources.
Online prediction	Interactive decision or application experience with strict latency.	Multi-zone managed endpoint where supported, autoscaling, health checks, timeout, retry policy and fallback.	Minimum replicas only when justified; monitor utilization and cost per successful prediction.
Streaming enrichment	Event-by-event or micro-batch scoring in a streaming pipeline.	Versioned model, checkpoint/replay, deterministic event ID and dead-letter handling.	Balance endpoint calls versus in-pipeline model footprint and batchability.
Asynchronous agent task	Long-running research/workflow beyond request timeout.	Durable task state, idempotent tool calls, progress status and human escalation.	Set token/tool/time budgets and cancellation.

11.1 Endpoint design standards

- Authenticate every call and propagate tenant/business purpose in trusted context.
- Define maximum request size, concurrency, timeout, retry and circuit-breaker behavior.
- Use traffic splitting or explicit aliases for canary and rollback.
- Do not log raw sensitive features or prompts by default.
- Test cold-start, autoscaling, regional dependency failure and quota exhaustion.
- Provide deterministic fallback: cached result, rules, prior model, manual process or graceful abstention.

12. Vertex AI Pipelines, ML Metadata and Experiments

Pipelines are the production authority for repeatable training, evaluation, registration and deployment. Notebook execution is permitted for exploration but is not production evidence unless converted into versioned components and rerun by the controlled pipeline identity.

Capability	Required use	Implementation standard
Vertex AI Pipelines	Orchestrate production ML and selected GenAI data/evaluation workflows.	Compiled KFP v2 template, immutable artifact, parameter contract and environment promotion.
ML Metadata	Capture execution/artifact lineage and relationships.	Link run to data snapshot, code, container, metrics, model and deployment.
Experiments	Compare runs, parameters and metrics during development.	Naming by use case/environment, tags, owner and retention.
Pipeline components	Reusable training, validation, bias, security and release gates.	Golden components with pinned versions and least-privilege runtime identity.
CI/CD	Compile, scan, test and promote pipeline templates.	No direct production submission from user notebooks.

12.1 Pipeline control flow

- Validate data contract and point-in-time snapshot.
- Compute/profile features and block material quality failures.
- Train with declared hardware, dependencies and parameters.
- Evaluate quality, segments, robustness, fairness, explainability, latency and economics.
- Run security and policy gates by risk tier.
- Register immutable artifact and evidence.
- Require approval before production alias/endpoint change.
- Verify deployment and automatically revert on failed health gate.

```
# Representative submission contract (illustrative shell)
python -m kfp.compiler.compiler_main   --py model_pipeline.py --function churn_pipeline --output pipeline.json

gcloud ai pipelines run   --region="${REGION}"   --display-name="churn-${GIT_SHA}"   --service-account="$
{PIPELINE_SA}"   --template-file=pipeline.json   --parameter-values="project=${PROJECT_ID},location=${REGION}"
```

13. Model Monitoring, Drift and Bias

Monitoring distinguishes service health from model behavior. Infrastructure availability can be healthy while predictions are stale, skewed, unfair or no longer useful. The platform therefore combines endpoint/job telemetry, feature/prediction distributions, delayed outcome quality, segment performance, business impact and incident feedback.

Signal	Example SLI	Action
Training-serving skew	Distribution distance for selected features.	Investigate pipeline/serving transformation mismatch; stop promotion if material.
Feature drift	Population shift from reference window.	Assess seasonality, source change or retraining need.
Prediction drift	Score/class distribution change.	Review input mix, model behavior and thresholds.
Quality drift	AUC, precision, calibration, forecast error or business outcome.	Retrain, adjust threshold, route to prior model or manual process.
Bias/segment drift	Performance or error-rate gap by approved segment.	Responsible-AI review, mitigation and potential suspension.
Data integrity	Missingness, range, category and freshness violations.	Block batch, degrade endpoint or quarantine input.
Operational	Latency, errors, throughput, resource saturation.	Scale, fail over, shed load or activate fallback.

13.1 Model Monitoring versions and validation

Google Cloud monitoring capabilities evolve and may differ by model type, endpoint, region and release maturity. The implementation team must validate whether the required skew, drift, attribution and generative monitoring capabilities are generally available for the selected workload. Preview-only capability cannot be the sole control for a regulated production obligation without an approved compensating control.

13.2 Bias monitoring

Protected attributes are processed only where legally and ethically approved for fairness testing. Access is restricted, results are aggregated where possible, and fairness metrics are selected based on harm model rather than a universal threshold. Monitoring includes intersectional segments when sample size and privacy permit.

14. Prompt Engineering, Management and Evaluation

Prompts are versioned software and policy artifacts. A production prompt package contains purpose, risk tier, system/developer instructions, input/output schemas, model alias, generation parameters, retrieval/tool contracts, guardrails, evaluation suite and owner. Prompt changes follow code review and evaluation; they are not edited directly in production consoles.

Prompt layer	Responsibility	Control
System policy	Non-negotiable role, safety, authorization and response behavior.	Central template, restricted modification and security review.
Developer workflow	Task procedure, tool use, evidence rules and output schema.	Application repository and code owner.
User input	Business request.	Length/type validation, injection detection and content classification.
Retrieved evidence	Authorized facts and context.	Delimit as untrusted data; preserve source IDs/ACL; ignore embedded instructions.
Tool result	Typed result from approved service.	Schema validation, provenance and error handling.
Output contract	Structured response, citations, confidence and action proposal.	Parser validation and human confirmation where required.

14.1 Evaluation hierarchy

Level	What is tested	Methods
Component	Prompt, retriever, reranker, tool or parser in isolation.	Unit cases, deterministic fixtures and schema tests.
End-to-end offline	Representative complete task.	Golden set, rubrics, model-based metrics calibrated with humans, safety suite.
Adversarial	Misuse, injection, exfiltration, unsafe and ambiguous inputs.	Red-team corpus, mutated prompts and tool-abuse tests.
Performance	Latency, concurrency, context size and quota behavior.	Load/soak tests and failure injection.
Online	Real behavior after release.	Feedback, sampled human review, quality proxies, safety/cost alerts and A/B/canary.

14.2 Evaluation release criteria

- Every critical metric has a threshold, confidence interval or explicit human acceptance rule.
- The evaluation dataset is versioned and contains representative, edge, stale/conflicting and adversarial cases.
- Model-based judges are calibrated against human labels and are not the sole validator for high-impact use cases.
- Regression comparison includes the currently deployed champion, not only an absolute score.
- Any model, prompt, retrieval, chunking, index, tool, safety or policy change triggers the relevant suite.

15. LLMOps Operating Lifecycle

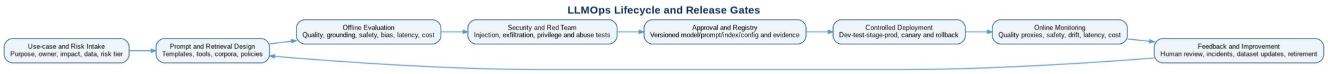

Figure 15-1. LLMOps lifecycle with risk intake, offline evaluation, red teaming, approval, controlled deployment and continuous improvement.

Artifact	Versioning unit	Promotion rule
Model	Approved alias resolving to provider model/version and region.	Change only after regression, safety, latency and cost evaluation.
Prompt package	Repository commit/tag and schema version.	Review, automated tests and owner approval.
Corpus/index	Source snapshot, chunking/embedding config and ACL metadata version.	Quality/security/deletion validation before alias switch.
Agent	Agent code/config, subagents, memory policy and tool catalog.	Trajectory, tool-security and final-response tests.
Tool	API/schema/version, scopes, timeout/retry and owner.	Contract/security test and least-privilege approval.
Evaluation	Dataset, rubric, judge and thresholds.	Independent control; changes reviewed to prevent metric gaming.
Runtime configuration	Region, quotas, cache, safety, observability and budgets.	IaC/config promotion and rollback.

15.1 GenAI change impact

Because behavior is emergent, a seemingly small configuration change can alter safety, tone, task success, citations or tool selection. The release manifest records the full resolved configuration. A model provider update, SDK change, corpus refresh or search backend change triggers targeted regression even when application code is unchanged.

15.2 Production feedback

User feedback is evidence, not truth. It is deduplicated, abuse-filtered, sampled and linked to trace/config versions. Confirmed failures become evaluation cases after privacy review. Teams are prohibited from optimizing solely for thumbs-up metrics when doing so could reduce accuracy, safety or fairness.

16. Enterprise RAG Architecture

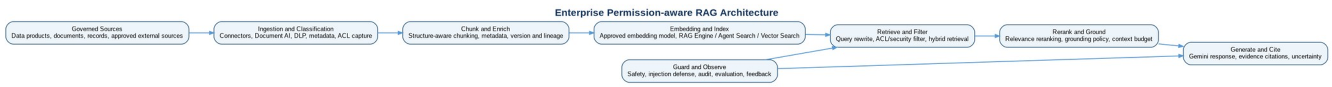

Figure 16-1. Permission-aware enterprise RAG from governed sources through retrieval, grounding, citations and guardrails.

RAG is selected when answers depend on changing enterprise knowledge, source evidence or authorization. It is not a substitute for deterministic transactions, trusted KPI calculations or legal interpretation. Structured facts should be retrieved from governed APIs/semantic models where possible; documents supply narrative evidence.

RAG pattern	Use when	Default service choice
Managed enterprise search	Rapid search/RAG over enterprise content with connectors, ranking and access features.	Agent Search, after connector/ACL/location assessment.
Managed corpus RAG	Application-controlled corpora and retrieval integrated with Gemini.	RAG Engine with an approved backend and location.
Dedicated vector retrieval	Large-scale custom embeddings, filtering and serving behavior.	Vector Search / Vector Search 2.0 subject to workload fit.
BigQuery vector search	Embeddings are co-located with governed analytical data and SQL joins/filtering are central.	BigQuery vector capabilities for suitable latency/scale.
Specialized external/open source	Required portability, algorithm or sovereign pattern not met by managed services.	Exception architecture with full operations/security ownership.

16.1 RAG trust contract

- Every chunk can be traced to source system, document, version, section, owner, classification and effective dates.
- Retrieval enforces the calling user/tenant entitlement before evidence reaches the model.
- The model is instructed to treat evidence as untrusted data and ignore embedded control instructions.
- Material factual claims include source citations or the system abstains.
- Deleted, expired or revoked source content is removed from retrieval within a defined SLO.
- Answer, evidence IDs, configuration version and policy decision are auditable subject to privacy minimization.

17. Document Ingestion, Chunking and Embeddings

Stage	Implementation pattern	Quality/security tests
Source registration	Register source owner, system of record, connector, region, classification, retention, legal hold and ACL model.	Authorization and records-management approval.
Ingestion	Event/schedule import with immutable source/version identifiers and idempotent job IDs.	Completeness, duplicate and deletion tests.
Extraction	Use native parsing or Document AI for layout/forms/OCR as justified.	Page/field confidence, table structure and language tests.
Classification	Sensitive Data Protection plus business metadata and policy.	False-negative sampling and exception workflow.
Chunking	Structure-aware sections with token target/overlap by content type.	Boundary coherence, context preservation and retrieval benchmark.
Embedding	Approved model/version, batch controls and source metadata.	Dimensional consistency, failed item reconciliation and re-embedding plan.
Index publication	Build immutable candidate then switch approved alias.	ACL filter, recall, performance and deletion validation.

17.1 Chunking guidance

Content type	Starting pattern	Do not
Policies/manuals	Heading-aware chunks with parent title and modest overlap.	Split obligations, conditions or exception clauses from their context.
Contracts/regulations	Clause/subclause chunks with document and effective-date metadata.	Merge jurisdictions or versions.
Tables/forms	Preserve row/column labels and page/form identifiers.	Flatten into ambiguous text without structure.
Tickets/cases	Separate events and summaries; redact/minimize personal data.	Index uncontrolled comments or attachments by default.
Code	Function/class/module-aware chunks with repository/commit metadata.	Mix branches or expose secrets.
Transcripts	Speaker/time segments plus topic boundaries.	Treat low-confidence speech recognition as authoritative.

17.2 Embedding governance

Embedding vectors are derived data and inherit source classification. The embedding model, dimensionality, normalization, language coverage and version are recorded. Model changes require re-embedding or compatibility proof; mixing incomparable vector spaces in one index is prohibited.

18. Vector Indexing, Retrieval and Reranking

Layer	Design decision	Acceptance criteria
Index topology	By region, tenant/domain, corpus risk and scale; avoid a universal unrestricted index.	Isolation and lifecycle align with source authorization.
Metadata filters	Tenant, user/group entitlement, classification, region, document status and effective date.	Negative cross-tenant/expired-content tests return zero unauthorized evidence.
Retrieval mode	Dense, sparse/keyword or hybrid based on benchmark.	Recall and precision meet domain baseline.
Query transformation	Spell, synonym, decomposition and safe rewrite.	Meaning preserved; injection/policy text not elevated.
Reranking	Use when candidate precision materially improves task quality.	Measured quality gain justifies latency/cost.
Context assembly	Deduplicate, diversify, order, budget and preserve citations.	No source flooding; critical evidence retained.
Cache	Cache safe query/retrieval results by entitlement and corpus version.	No cross-user leak; invalidation meets deletion/change SLO.

18.1 Backend selection

The vector backend is selected per workload using scale, latency, filtering, hybrid search, update/delete behavior, regional support, VPC Service Controls, CMEK, operational model and cost. Architecture teams must validate current RAG Engine backend constraints and location availability; a capability shown in preview or one location is not assumed globally.

19. Grounding, Citations and Answer Quality

Grounding means the response is supported by approved evidence, not merely plausible. Citations are generated from retrieved source identifiers after answer construction and are validated for entailment and coverage. A link alone is not evidence if it does not support the claim or the user cannot access it.

Control	Implementation	Failure behavior
Evidence-only mode	System policy requires claims from supplied evidence.	Abstain or state uncertainty.
Claim-evidence mapping	Associate material claims with source/chunk IDs.	Remove unsupported claim or request more evidence.
Citation validation	Check that cited passage supports claim and remains authorized.	Suppress citation/answer and log quality issue.
Conflicting evidence	Detect source/version disagreement and apply precedence rules.	Show conflict and escalate rather than fabricate resolution.
Staleness	Effective/expiry date filters and corpus freshness SLO.	Warn, abstain or route to source owner.
Structured truth	Use APIs/BigQuery/semantic metrics for precise facts.	Do not infer transactional/KPI values from prose.

19.1 Quality scorecard

Metric	Definition	Illustrative gate
Retrieval recall@K	Relevant evidence present in candidate set.	Domain-approved baseline; no material regression.
Groundedness	Material answer claims supported by evidence.	At least 90% for Tier 2 baseline; stricter by use case.
Citation precision	Citations actually support associated claims.	At least 95% baseline.
Abstention accuracy	Correct refusal when evidence/authorization is insufficient.	At least 95% on negative set.
Task success	Human/rubric outcome for representative workflow.	At least 85% or business-approved threshold.
Authorization leakage	Unauthorized evidence or answer exposure.	Zero critical cases.

20. Agent and Multi-agent Architecture

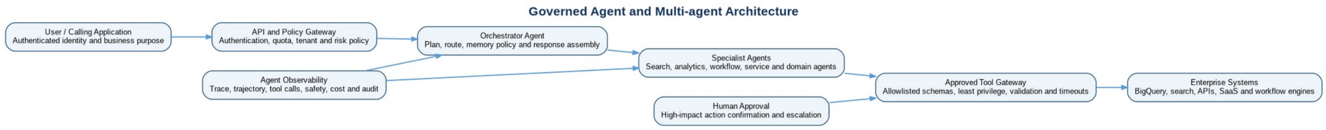

Figure 20-1. Governed agent architecture with authenticated entry, orchestrator, specialist agents, approved tool gateway, human approval and observability.

An agent is approved only when orchestration or tool use adds measurable value beyond a deterministic workflow or single model call. The design minimizes autonomy by default: plans are bounded, tools are allowlisted, arguments are validated, permissions are scoped to the runtime identity and consequential actions require confirmation or an explicit approved policy.

Component	Responsibility	Mandatory control
API/policy gateway	Authenticate caller, establish tenant/purpose/risk and enforce quotas.	Trusted identity context; reject client-supplied entitlement claims.
Orchestrator agent	Plan, route, manage bounded state and assemble response.	Maximum steps/time/cost; explicit allowed subagents and fallback.
Specialist agent	Execute one domain capability.	Narrow instruction, tools, corpus and identity.
Tool gateway	Expose approved business operations through typed interfaces.	Schema validation, least privilege, idempotency, timeout, audit and policy.
Memory	Store only approved conversational/task state.	Classification, minimization, TTL, tenant isolation and deletion.
Human approval	Confirm high-impact, irreversible or externally visible actions.	Show proposed action, evidence and consequences; prevent approval spoofing.
Observability	Capture trajectory, decisions, tool calls, outcomes and cost.	Redaction, access control and trace-to-release correlation.

20.1 Multi-agent design rules

- Use multiple agents only where domain separation, scaling, ownership or evaluation is clearer than one orchestrator with tools.
- Do not allow agents to dynamically grant permissions, create unrestricted tools or bypass the policy gateway.
- Every inter-agent message has a schema, source identity, task ID and classification.
- Prevent recursive delegation loops with depth, step and budget limits.
- Evaluate both trajectory and final response; a correct answer reached through an unsafe tool path fails.
- Maintain a deterministic emergency disable for each agent and tool.

20.2 Agent Development Kit and Runtime

Agent Development Kit is the preferred framework for new governed agents where its language/runtime support fits. Managed Agent Runtime is preferred for enterprise deployment where current regional, network, observability and security capabilities satisfy the workload. Cloud Run or another approved runtime remains an alternative for patterns needing more portability or controls not yet available in the managed runtime.

21. Document, Speech, Vision and Multimodal AI

Capability	Representative enterprise use	Architecture pattern	Primary risks
Document AI	Invoice, form, claims, contract and records extraction.	Processor -> confidence/validation -> structured governed product -> human exception.	OCR/extraction error, sensitive documents, processor location/version.
Speech-to-Text	Contact-center transcription, meeting/case capture and commands.	Streaming/batch transcription -> redaction -> quality -> downstream workflow.	Consent, speaker identity, low-confidence facts and retention.
Text-to-Speech	Accessible interfaces and approved voice experiences.	Text policy -> synthesis -> channel controls -> disclosure.	Impersonation, brand/voice rights and unsafe content.
Vision AI	Image classification, object/text detection and inspection.	Approved model/API -> confidence -> review -> event/product.	Bias, surveillance, biometric/legal constraints and false detection.
Multimodal Gemini	Reason across text, image, audio/video and documents.	Input policy -> model -> structured output/evidence -> review.	Hidden sensitive content, oversized context and cross-modal hallucination.

21.1 Confidence and human exception handling

Specialized extraction/classification outputs use field- or object-level confidence and business validation. Low-confidence or high-value records are routed to human review. Corrections are retained as labeled evidence only after privacy and quality checks; they do not automatically retrain a model.

21.2 Synthetic media controls

- Disclose AI-generated customer-facing audio/image/video where policy or law requires.
- Restrict voice cloning, biometric inference and identity manipulation.
- Preserve provenance/watermark metadata where supported and meaningful.
- Require brand, legal and security review for public synthetic media.

22. AI Security and Model Access Controls

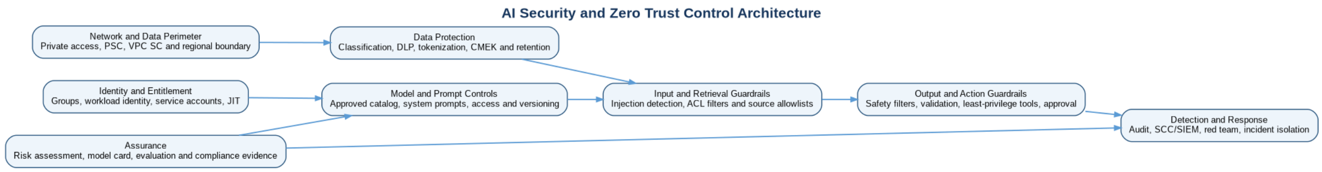

Figure 22-1. Defense-in-depth AI security across identity, network, data, models/prompts, retrieval, output/actions, detection and assurance.

Security boundary	Mandatory controls
Human identity	Enterprise SSO/MFA, group lifecycle, context-aware access, JIT privileged access and quarterly review.
Workload identity	Dedicated service account per runtime/pipeline, Workload Identity Federation, no service-account keys and least privilege.
Model access	Approved catalog/aliases, allow by project/group/service account, quotas and deny unapproved external endpoints.
Network	Private Google Access/PSC where supported, controlled egress, DNS/firewall policy and VPC Service Controls for sensitive services.
Data	Classification, minimization, authorized products, DLP/tokenization, encryption/CMEK where supported, retention and deletion.
Secrets	Secret Manager, rotation, no prompts/config/code secrets and tool-specific credentials.
Supply chain	Pinned dependencies, signed/scanned containers, provenance and controlled pipeline templates.
Audit	Admin/data access/model/pipeline/agent/tool events centralized with tamper-resistant retention.

22.1 Model access policy

Risk/data class	Permitted model endpoint	External grounding	Training/fine-tuning
Public/Internal Tier 1	Approved enterprise model alias.	Conditional with approved source policy.	Only approved datasets and provider data-use terms.
Confidential Tier 2	Approved regional endpoint with enterprise controls.	Default deny; explicit domain/security approval.	Authorized data product, minimization and lineage.
Restricted/Regulated Tier 2-3	Validated location, private/perimeter and encryption controls.	Prohibited by default.	Case-specific legal/privacy/security approval and evidence.
Prohibited Tier 4	No production access.	No.	No.

22.2 Sensitive Data Protection

Sensitive Data Protection is used to discover/classify selected inputs, corpora and outputs; de-identification or tokenization is applied where the business purpose permits. DLP scanning is not treated as proof that content is safe: custom identifiers, context, attachments, images and multilingual content require workload-specific validation.

23. Prompt-injection and Data-exfiltration Protection

Prompt injection is treated as an authorization and control-plane attack, not solely a content-moderation problem. Retrieved documents, web pages, emails, tool output and user input are untrusted. No text instruction can grant permissions or override system/tool policy.

Threat	Preventive controls	Detective/response controls
Direct injection	Instruction hierarchy, input segmentation, policy classifier and restricted capabilities.	Adversarial tests, policy-violation trace and user/rate action.
Indirect injection in documents/web	Sanitize/segment content; tell model evidence is data; source allowlist and retrieval metadata.	Flag suspicious chunks, suppress source, re-index and investigate source compromise.
Tool abuse	Typed narrow tools, server-side entitlement, argument validation, action policy and human confirmation.	Tool-call anomaly, rejected action metrics and immediate tool disable.
Data exfiltration	Least privilege, tenant filters, output DLP, egress allowlist and secret isolation.	Canary data, audit correlation, incident isolation and credential rotation.
Cross-tenant memory/cache leak	Tenant-keyed memory/cache, TTL, encryption and no shared implicit context.	Negative tests and cache/memory purge.
Model extraction/abuse	Quota, anomaly detection, response limits and no sensitive training by default.	Rate limiting, block identity and provider escalation.

23.1 Safe tool-use pattern

- The agent proposes an action using a typed schema; a policy service independently authorizes it.
- The tool service rechecks caller/runtime identity, tenant and business rule—never trusting the model.
- High-impact action previews show target, values, evidence and reversibility to a human approver.
- Mutating calls carry idempotency keys and write an immutable audit event.
- Tool output is minimized and returned in a typed envelope; errors do not expose secrets or internal stack traces.

Prohibited anti-pattern. Giving an agent a general shell, browser, database-owner credential or unrestricted HTTP tool in production and relying on the prompt to keep it safe is prohibited.

24. Responsible AI, Human Oversight and Audit

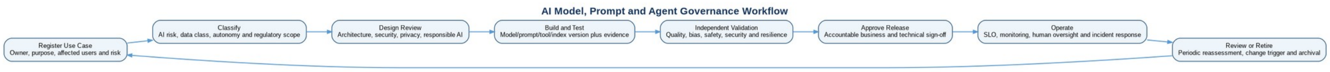

Figure 24-1. Risk-based governance workflow from registration and classification through independent validation, approval, operation and retirement.

Responsible-AI principle	Implementation control	Evidence
Purpose and accountability	Named business owner, intended/prohibited use and impact assessment.	Use-case record and approval.
Fairness	Relevant segment analysis, data representativeness, mitigation and periodic review.	Fairness report and decision log.
Transparency	User disclosure, limitations, citations/explanations appropriate to context.	UX review and model/prompt card.
Privacy	Minimization, lawful basis, consent where needed, retention and deletion.	Privacy assessment and data lineage.
Safety and robustness	Misuse tests, red team, safe failure, human escalation and incident plan.	Evaluation/security evidence.
Human agency	Review/confirmation, appeal/correction and no deceptive automation.	Workflow and audit trail.
Accountability/audit	Immutable release/config evidence and traceable decisions/actions.	Release package, logs and review records.

24.1 Human oversight patterns

Pattern	Use	Design requirement
Human-in-the-loop	Every high-impact output/action reviewed before execution.	Reviewer has context, evidence, competence, time and genuine authority to reject.
Human-on-the-loop	Automated operation with active supervision and intervention.	Real-time monitoring, stop control, bounded scope and rapid escalation.
Human-in-command	Human defines policy/goals and reviews system performance periodically.	Suitable only for lower-impact bounded automation with reliable controls.
Manual fallback	AI unavailable/uncertain; process continues without it.	Document capacity, data handoff and recovery.

24.2 AI audit record

- Use-case, owner, risk tier and approved purpose.
- Caller/runtime identity and tenant context.
- Resolved model, prompt, retrieval/index, tool and policy versions.
- Input/output hashes or minimized content as permitted; evidence and citations.
- Safety/policy decisions, human approval and tool/action outcome.
- Latency, tokens/resources, errors and trace ID.
- Retention/classification and access history.

25. AI Observability, SLOs and FinOps

Observability correlates user experience, model behavior, retrieval, tools, infrastructure and economics. Telemetry is designed before release and minimized because prompts, evidence, tool arguments and outputs can contain sensitive data.

Layer	Key signals
Experience	Successful task, abandonment, escalation, user feedback and business outcome.
Generation	Model/alias, token counts, latency, finish reason, safety decision and schema validity.
RAG	Query, retrieval latency, candidate count, filters, source coverage, grounding and citation metrics.
Agent	Trajectory, steps, selected tools, rejected calls, loops, human approvals and action outcome.
ML	Endpoint/job availability, latency, errors, feature/prediction drift and delayed quality.
Data/knowledge	Freshness, quality, corpus ingestion, ACL/deletion lag and index version.
Security	Injection/policy events, exfiltration indicators, entitlement failures and anomalous access.
FinOps	Cost by domain/use case/model/task; tokens, cache, vector/search, endpoint, accelerator and tool/API cost.

25.1 SLO catalog

Use case	Availability/success	Performance	Quality/safety	Recovery
Tier 1 online prediction	99.9% successful endpoint responses.	P95 per product target.	Drift/quality within approved bounds.	RTO <4h, RPO <15m configuration; model reconstructable.
Tier 1 RAG assistant	99.9% successful bounded responses.	P95 end-to-end and retrieval targets.	Grounding/citation and zero critical authorization leakage.	RTO <4h; corpus/index recovery per source tier.
Tier 0 agent workflow	99.99% or business-approved with deterministic fallback.	Step and total-task deadlines.	Zero unauthorized action; human approval reliability.	RTO <1h, RPO <5m for durable task state if required.
Batch scoring	99% completion by deadline.	Throughput/cost envelope.	Reconciliation and quality gates.	Rerun from immutable input.

25.2 Unit economics

The primary metric is cost per successful business task or prediction, decomposed into model input/output tokens, context cache, retrieval/search/vector operations, embeddings, endpoint/accelerator time, pipeline compute, storage, network and tool/API calls. Forecasts use current measurable variables entered into the Google Cloud Pricing Calculator; this volume does not invent current prices.

Optimization lever	Guardrail
Smaller/faster model for simple tasks	Must pass the same task-specific quality and safety thresholds.
Reduce prompt/context and improve retrieval	Do not remove necessary policy/evidence or degrade grounding.
Caching	Key by authorization and version; enforce invalidation and privacy.
Batching/offline processing	Use only where latency and data-protection requirements permit.
Endpoint scaling/accelerators	Load-test; avoid idle minimums or oversized hardware without SLO evidence.
Limit agent steps/tools	Maintain task success and safe fallback.

26. AI Disaster Recovery and Resilience

Component	Backup/reconstruction source	Recovery pattern
Model artifact/registry metadata	Immutable artifact registry/model source, pipeline and evidence.	Re-register/deploy prior approved version in recovery region after compatibility validation.
Prompt/agent/tool config	Git and immutable release manifest.	Redeploy prior release and disable risky tools.
Training/evaluation data	Versioned governed data products/snapshots subject to retention.	Recreate by pipeline or restore approved snapshot.
RAG corpus/index	Source-of-record documents, metadata/ACL export and ingestion config.	Rebuild index; use prior index alias or degraded source search while rebuilding.
Feature data	BigQuery history/source and feature transformation code.	Recompute/offline restore; online store recovered according to tier.
Endpoint/runtime	IaC/config and container/model artifacts.	Deploy to recovery location/project; update routing after security and smoke test.
Durable agent task state	Regional durable store/event log where business process requires.	Replay idempotently and reconcile external actions.
Audit/evidence	Central protected logging/evidence repository.	Restore access and preserve chain of custody.

26.1 Failure scenarios

Scenario	Expected behavior
Model/API quota or regional outage	Circuit breaker, alternate approved model/location if authorized, or deterministic/manual fallback.
Corrupt/new corpus index	Freeze alias, route to prior approved index, stop ingestion and rebuild from source.
Unsafe model behavior after provider change	Disable alias/version, revert prompt/model configuration and notify governance/security.
Tool/backend unavailable	Agent returns bounded partial result or queues task; no repeated unsafe writes.
Telemetry unavailable	High-risk actions fail closed or enter degraded mode; restore audit before normal operation.
Compromised credential/tool	Revoke identity, disable tool/agent, rotate secret, isolate project and investigate audit.

DR exercises include model endpoint restoration, corpus/index rebuild, agent/tool disable, release rollback, source-deletion propagation and a tabletop for unsafe or unauthorized behavior. Recovery objectives are workload-specific and cannot exceed underlying service/data dependencies.

27. AI Governance and Operating Model

Role	Accountability
Enterprise AI Governance Council	Policy, risk tiers, prohibited uses, exception authority, portfolio assurance and periodic review.
AI Platform Product Team	Managed platform, golden components, approved model catalog, runtime patterns, observability and developer experience.
Domain AI Product Owner	Business outcome, use-case lifecycle, budget, SLO, users and domain approvals.
Data/Knowledge Owner	Authorization, quality, lineage, source precedence, ACL and deletion behavior.
ML/GenAI/Agent Engineering	Design, build, test, documentation, deployment and defect remediation.
Independent Validation / Responsible AI	Challenge evaluation, fairness, safety, robustness and limitations.
Security and Privacy	Threat model, access, network, DLP, incident and regulatory privacy controls.
SRE and FinOps	Reliability, on-call, capacity, unit economics and optimization.
Legal/Compliance/Records	Applicable obligations, disclosure, retention, legal hold and audit evidence.

27.1 Governance forums and cadence

Forum	Cadence	Decisions/evidence
Use-case intake	Weekly	Purpose, risk tier, ownership, data and architecture path.
Model/prompt release review	Per release for Tier 2/3	Evaluation, security, cost, rollback and approval.
Operational review	Weekly/monthly by tier	SLO, incidents, quality/safety, cost and feedback.
Responsible-AI review	Quarterly and event-driven	Segment outcomes, harms, complaints, drift and controls.
Portfolio/value review	Quarterly	Benefit, adoption, unit economics, duplicate solutions and retirement.
Architecture/product validation	Quarterly	Current services, regions, quotas, VPC SC, CMEK, SDK/provider and commercial changes.

27.2 Model governance workflow

- Register and classify use case.
- Complete architecture, threat, privacy and responsible-AI assessments by tier.
- Build immutable model/prompt/index/agent artifacts and evidence.
- Run independent validation where required.
- Approve and promote through separated duties.
- Operate with SLOs, human oversight, monitoring, incidents and cost review.
- Reassess after material change and retire when value, safety or support no longer justifies operation.

28. Structured Service Assessments

Each selected Google Cloud service maps to a documented capability and is assessed using the master blueprint's 32-field structure. Quotas, regional availability, pricing, VPC Service Controls, CMEK, SDK and Terraform support must be validated against current official documentation before implementation; no static quota or price is asserted here.

28.1 Gemini Enterprise Agent Platform / Vertex AI

Assessment field	Guidance
Service name	Gemini Enterprise Agent Platform / Vertex AI
Purpose	Managed enterprise AI platform for generative models, predictive ML, pipelines, model operations, RAG and agents.
Enterprise architecture role	Primary AI control and execution plane integrated with inherited data, identity, network, governance and operations.
Business capabilities supported	AI product delivery, decision support, enterprise knowledge access, workflow automation, risk control and operational efficiency.
Technical capabilities	Model access and tuning; custom/AutoML training; batch/online prediction; pipelines; registry; evaluation; RAG; agents.
Benefits	Unified managed platform, Google Cloud integration, enterprise controls and reduced self-managed serving/operations.
Limitations	Feature maturity, regional availability and control support differ by capability; product terminology and SDKs evolve.
Service quotas	Validate current per-project, per-region, request, token, job, storage and concurrency quotas; establish quota owners, alerts and increase lead time.
Regional availability considerations	Validate the exact capability/model/version in AMER, EMEA and APAC target locations; co-locate data, runtime and telemetry and prohibit unapproved cross-region movement.
High-availability characteristics	Managed regional/multi-zone characteristics depend on capability; applications still require retries, fallback and regional recovery.
Disaster-recovery considerations	Reconstruct from IaC, artifacts, pipelines, model/prompt/index manifests and source data; validate cross-region model/capability availability.
Security model	IAM/service agents, workload identities, project/location boundaries, VPC SC/private access/CMEK where supported, audit and safety controls.
IAM requirements	Use least-privilege predefined/custom roles, dedicated workload identities, service-agent review, separation of builder/approver/deployer and no service-account keys.
Network requirements	Use private Google access/PSC and controlled egress where supported; co-locate with regional data.
VPC Service Controls considerations	Validate current inclusion, ingress/egress methods, supported APIs and dependent services; test perimeter paths and document exceptions.
Encryption capabilities	Google-managed encryption by default; validate in-transit protection and data/artifact encryption for each sub-capability.
CMEK support	Validate current resource/model/location-specific CMEK support and key lifecycle; do not claim unsupported coverage.
Logging	Enable required Admin Activity/Data Access/service logs and application correlation; minimize/redact sensitive prompts, features, evidence and outputs.
Monitoring	Monitor availability, errors, latency, throughput, quality/safety signals, quota, data freshness and unit economics.
Alerting	Route severity by risk tier to product/SRE/security; alert on telemetry absence, authorization/safety violations, SLO burn and budget anomalies.
Cost drivers	Model tokens, training/prediction compute, endpoints, pipelines, storage, networking and related managed capabilities.
Cost-optimization techniques	Measure cost per successful task/prediction; select fit-for-purpose model/resources, bound context/steps, batch/cache safely, co-locate and retire idle artifacts.
Operational responsibilities	AI platform owns shared service patterns; domain product owns use case/data/quality; SRE owns reliability; Security/Privacy/Responsible AI own assurance controls.
Integration points	BigQuery, Cloud Storage, Knowledge Catalog, IAM, KMS, Logging/Monitoring, CI/CD, Artifact Registry, domain APIs and applicable AI services.
Alternatives	Self-managed models on GKE/Cloud Run; third-party AI platforms; point services.
Selection criteria	Use as default when managed capability, security/location support and workload economics meet requirements.
Best practices	Use approved templates, immutable versions, regional isolation, risk-tier evidence, automated tests, least privilege, SLOs, cost budgets and tested rollback.
Reference architecture	Primary AI control and execution plane integrated with inherited data, identity, network, governance and operations. Integrated through the enterprise AI platform, trust/control plane and regional data products.
Sample configuration	Regional AI projects, dedicated runtime/pipeline service accounts, approved model aliases, centralized audit and Terraform-managed foundations.
Common pitfalls	Treating the platform as one undifferentiated service or assuming every sub-capability has identical location/security support.
Anti-patterns	Unmanaged console-only production, embedded credentials, raw/unclassified data, hard-coded mutable model identifiers, missing evaluation or no rollback.
Production-readiness checklist	Current product/location/quota/security validation; approved use case and data; IAM/network/KMS/logging; tests/evaluation; SLO/alerts/runbook; capacity/cost; DR and rollback evidence.

28.2 Gemini models and Model Garden

Assessment field	Guidance
Service name	Gemini models and Model Garden
Purpose	Provide approved foundation, embedding and specialized models for generative and AI workloads.
Enterprise architecture role	Curated enterprise model catalog behind stable aliases and use-case evaluation gates.
Business capabilities supported	AI product delivery, decision support, enterprise knowledge access, workflow automation, risk control and operational efficiency.
Technical capabilities	Text/code/multimodal generation, embeddings, model variants, tuning options and provider-model discovery.
Benefits	Rapid access to capable models with managed inference and a catalog of alternatives.
Limitations	Non-deterministic behavior; model availability/versioning, context, safety and data terms vary.
Service quotas	Validate current per-project, per-region, request, token, job, storage and concurrency quotas; establish quota owners, alerts and increase lead time.
Regional availability considerations	Validate the exact capability/model/version in AMER, EMEA and APAC target locations; co-locate data, runtime and telemetry and prohibit unapproved cross-region movement.
High-availability characteristics	Managed endpoint characteristics; clients use timeout, retry, quotas, circuit breaker and fallback.
Disaster-recovery considerations	Maintain approved alternative alias/model/location and reproducible prompt/evaluation package.
Security model	Restrict model usage by project/identity; validate provider/data-use terms; prevent secrets and unauthorized context.
IAM requirements	Use least-privilege predefined/custom roles, dedicated workload identities, service-agent review, separation of builder/approver/deployer and no service-account keys.
Network requirements	Call from approved regional runtime with private/perimeter controls where supported.
VPC Service Controls considerations	Validate current inclusion, ingress/egress methods, supported APIs and dependent services; test perimeter paths and document exceptions.
Encryption capabilities	Google-managed encryption by default; validate in-transit protection and data/artifact encryption for each sub-capability.
CMEK support	Validate current resource/model/location-specific CMEK support and key lifecycle; do not claim unsupported coverage.
Logging	Enable required Admin Activity/Data Access/service logs and application correlation; minimize/redact sensitive prompts, features, evidence and outputs.
Monitoring	Monitor availability, errors, latency, throughput, quality/safety signals, quota, data freshness and unit economics.
Alerting	Route severity by risk tier to product/SRE/security; alert on telemetry absence, authorization/safety violations, SLO burn and budget anomalies.
Cost drivers	Input/output/cached tokens, provisioned capacity if chosen, tuning and evaluation calls.
Cost-optimization techniques	Measure cost per successful task/prediction; select fit-for-purpose model/resources, bound context/steps, batch/cache safely, co-locate and retire idle artifacts.
Operational responsibilities	AI platform owns shared service patterns; domain product owns use case/data/quality; SRE owns reliability; Security/Privacy/Responsible AI own assurance controls.
Integration points	BigQuery, Cloud Storage, Knowledge Catalog, IAM, KMS, Logging/Monitoring, CI/CD, Artifact Registry, domain APIs and applicable AI services.
Alternatives	Third-party hosted models; open models; specialized predictive/NLP services.
Selection criteria	Task quality, safety, location, latency, cost, context, multimodality, support and portability.
Best practices	Use approved templates, immutable versions, regional isolation, risk-tier evidence, automated tests, least privilege, SLOs, cost budgets and tested rollback.
Reference architecture	Curated enterprise model catalog behind stable aliases and use-case evaluation gates. Integrated through the enterprise AI platform, trust/control plane and regional data products.
Sample configuration	Alias gemini-approved-balanced resolves to tested model/config; application records resolved version per trace.
Common pitfalls	Hard-coding “latest” model IDs or promoting a model based on generic benchmarks rather than enterprise tasks.
Anti-patterns	Unmanaged console-only production, embedded credentials, raw/unclassified data, hard-coded mutable model identifiers, missing evaluation or no rollback.
Production-readiness checklist	Current product/location/quota/security validation; approved use case and data; IAM/network/KMS/logging; tests/evaluation; SLO/alerts/runbook; capacity/cost; DR and rollback evidence.

28.3 Vertex AI Pipelines

Assessment field	Guidance
Service name	Vertex AI Pipelines
Purpose	Managed orchestration for reproducible ML and selected GenAI lifecycle workflows.
Enterprise architecture role	Production authority for training, evaluation, registration, index build and controlled release steps.
Business capabilities supported	AI product delivery, decision support, enterprise knowledge access, workflow automation, risk control and operational efficiency.
Technical capabilities	KFP v2 pipeline execution, managed components, scheduling, parameterization and metadata integration.
Benefits	Repeatability, managed execution, lineage integration and component reuse.
Limitations	Component/API version compatibility, pipeline design complexity and regional/service dependencies.
Service quotas	Validate current per-project, per-region, request, token, job, storage and concurrency quotas; establish quota owners, alerts and increase lead time.
Regional availability considerations	Validate the exact capability/model/version in AMER, EMEA and APAC target locations; co-locate data, runtime and telemetry and prohibit unapproved cross-region movement.
High-availability characteristics	Managed execution; design idempotent components, retries and resumable artifacts.
Disaster-recovery considerations	Store compiled templates, source, containers and artifacts; resubmit in approved recovery project/location.
Security model	Dedicated pipeline service accounts, protected root, least privilege, private access and no secrets in parameters.
IAM requirements	Use least-privilege predefined/custom roles, dedicated workload identities, service-agent review, separation of builder/approver/deployer and no service-account keys.
Network requirements	Private data/resource connectivity and egress controls as workload requires.
VPC Service Controls considerations	Validate current inclusion, ingress/egress methods, supported APIs and dependent services; test perimeter paths and document exceptions.
Encryption capabilities	Google-managed encryption by default; validate in-transit protection and data/artifact encryption for each sub-capability.
CMEK support	Validate current resource/model/location-specific CMEK support and key lifecycle; do not claim unsupported coverage.
Logging	Enable required Admin Activity/Data Access/service logs and application correlation; minimize/redact sensitive prompts, features, evidence and outputs.
Monitoring	Monitor availability, errors, latency, throughput, quality/safety signals, quota, data freshness and unit economics.
Alerting	Route severity by risk tier to product/SRE/security; alert on telemetry absence, authorization/safety violations, SLO burn and budget anomalies.
Cost drivers	Component compute/training, pipeline execution dependencies, storage and logs.
Cost-optimization techniques	Measure cost per successful task/prediction; select fit-for-purpose model/resources, bound context/steps, batch/cache safely, co-locate and retire idle artifacts.
Operational responsibilities	AI platform owns shared service patterns; domain product owns use case/data/quality; SRE owns reliability; Security/Privacy/Responsible AI own assurance controls.
Integration points	BigQuery, Cloud Storage, Knowledge Catalog, IAM, KMS, Logging/Monitoring, CI/CD, Artifact Registry, domain APIs and applicable AI services.
Alternatives	Cloud Composer, Workflows, external KFP/Argo, CI orchestrators.
Selection criteria	Use when artifact lineage, ML components and managed orchestration outweigh generic workflow needs.
Best practices	Use approved templates, immutable versions, regional isolation, risk-tier evidence, automated tests, least privilege, SLOs, cost budgets and tested rollback.
Reference architecture	Production authority for training, evaluation, registration, index build and controlled release steps. Integrated through the enterprise AI platform, trust/control plane and regional data products.
Sample configuration	Compiled KFP v2 template with data gate, custom training, evaluation, registry and approval boundary.
Common pitfalls	Using notebooks as production pipelines or embedding mutable dependencies and user credentials.
Anti-patterns	Unmanaged console-only production, embedded credentials, raw/unclassified data, hard-coded mutable model identifiers, missing evaluation or no rollback.
Production-readiness checklist	Current product/location/quota/security validation; approved use case and data; IAM/network/KMS/logging; tests/evaluation; SLO/alerts/runbook; capacity/cost; DR and rollback evidence.

28.4 Vertex AI Model Registry

Assessment field	Guidance
Service name	Vertex AI Model Registry
Purpose	Catalog and version models with deployment and lineage metadata.
Enterprise architecture role	Authoritative inventory of production-eligible model versions and aliases.
Business capabilities supported	AI product delivery, decision support, enterprise knowledge access, workflow automation, risk control and operational efficiency.
Technical capabilities	Model versions, aliases, metadata, deployment relationships and import/upload patterns.
Benefits	Traceability, reuse, controlled promotion and rollback to approved versions.
Limitations	Does not by itself approve quality, fairness, security or business use; metadata discipline is required.
Service quotas	Validate current per-project, per-region, request, token, job, storage and concurrency quotas; establish quota owners, alerts and increase lead time.
Regional availability considerations	Validate the exact capability/model/version in AMER, EMEA and APAC target locations; co-locate data, runtime and telemetry and prohibit unapproved cross-region movement.
High-availability characteristics	Managed metadata service; retain independent source artifacts and release evidence.
Disaster-recovery considerations	Recreate entries from immutable model artifacts, manifests, pipelines and evidence if required.
Security model	Separate model uploader, approver and deployer roles; restrict artifact/model access.
IAM requirements	Use least-privilege predefined/custom roles, dedicated workload identities, service-agent review, separation of builder/approver/deployer and no service-account keys.
Network requirements	Access through approved APIs and project boundaries; artifacts co-located where required.
VPC Service Controls considerations	Validate current inclusion, ingress/egress methods, supported APIs and dependent services; test perimeter paths and document exceptions.
Encryption capabilities	Google-managed encryption by default; validate in-transit protection and data/artifact encryption for each sub-capability.
CMEK support	Validate current resource/model/location-specific CMEK support and key lifecycle; do not claim unsupported coverage.
Logging	Enable required Admin Activity/Data Access/service logs and application correlation; minimize/redact sensitive prompts, features, evidence and outputs.
Monitoring	Monitor availability, errors, latency, throughput, quality/safety signals, quota, data freshness and unit economics.
Alerting	Route severity by risk tier to product/SRE/security; alert on telemetry absence, authorization/safety violations, SLO burn and budget anomalies.
Cost drivers	Stored artifacts and downstream serving/training; registry metadata is not the main cost driver.
Cost-optimization techniques	Measure cost per successful task/prediction; select fit-for-purpose model/resources, bound context/steps, batch/cache safely, co-locate and retire idle artifacts.
Operational responsibilities	AI platform owns shared service patterns; domain product owns use case/data/quality; SRE owns reliability; Security/Privacy/Responsible AI own assurance controls.
Integration points	BigQuery, Cloud Storage, Knowledge Catalog, IAM, KMS, Logging/Monitoring, CI/CD, Artifact Registry, domain APIs and applicable AI services.
Alternatives	Artifact Registry/MLflow/custom registry; third-party MLOps catalogs.
Selection criteria	Default for Vertex AI model lifecycle and deployment integration.
Best practices	Use approved templates, immutable versions, regional isolation, risk-tier evidence, automated tests, least privilege, SLOs, cost budgets and tested rollback.
Reference architecture	Authoritative inventory of production-eligible model versions and aliases. Integrated through the enterprise AI platform, trust/control plane and regional data products.
Sample configuration	Immutable version 3.2.0 with aliases candidate/champion, model card URI, pipeline run and approval ID.
Common pitfalls	Overwriting mutable artifacts or treating an upload as equivalent to production approval.
Anti-patterns	Unmanaged console-only production, embedded credentials, raw/unclassified data, hard-coded mutable model identifiers, missing evaluation or no rollback.
Production-readiness checklist	Current product/location/quota/security validation; approved use case and data; IAM/network/KMS/logging; tests/evaluation; SLO/alerts/runbook; capacity/cost; DR and rollback evidence.

28.5 Vertex AI Experiments and ML Metadata

Assessment field	Guidance
Service name	Vertex AI Experiments and ML Metadata
Purpose	Track development runs, parameters, metrics, artifacts and lineage.
Enterprise architecture role	Evidence layer connecting data, code, pipeline execution, model and deployment.
Business capabilities supported	AI product delivery, decision support, enterprise knowledge access, workflow automation, risk control and operational efficiency.
Technical capabilities	Experiment/run comparison, metadata artifacts/executions/contexts and lineage visualization/integration.
Benefits	Reproducibility, impact analysis, audit and faster development comparison.
Limitations	Requires consistent instrumentation/naming; does not replace source control or governance approval.
Service quotas	Validate current per-project, per-region, request, token, job, storage and concurrency quotas; establish quota owners, alerts and increase lead time.
Regional availability considerations	Validate the exact capability/model/version in AMER, EMEA and APAC target locations; co-locate data, runtime and telemetry and prohibit unapproved cross-region movement.
High-availability characteristics	Managed metadata; critical evidence is also exported/retained per records policy.
Disaster-recovery considerations	Reconstruct from pipeline logs/artifacts and protected evidence export where feasible.
Security model	Restrict metadata because parameters, URIs and labels can expose sensitive context.
IAM requirements	Use least-privilege predefined/custom roles, dedicated workload identities, service-agent review, separation of builder/approver/deployer and no service-account keys.
Network requirements	Regional API access aligned with pipeline/project.
VPC Service Controls considerations	Validate current inclusion, ingress/egress methods, supported APIs and dependent services; test perimeter paths and document exceptions.
Encryption capabilities	Google-managed encryption by default; validate in-transit protection and data/artifact encryption for each sub-capability.
CMEK support	Validate current resource/model/location-specific CMEK support and key lifecycle; do not claim unsupported coverage.
Logging	Enable required Admin Activity/Data Access/service logs and application correlation; minimize/redact sensitive prompts, features, evidence and outputs.
Monitoring	Monitor availability, errors, latency, throughput, quality/safety signals, quota, data freshness and unit economics.
Alerting	Route severity by risk tier to product/SRE/security; alert on telemetry absence, authorization/safety violations, SLO burn and budget anomalies.
Cost drivers	Metadata operations, storage/logging and experiment compute executed elsewhere.
Cost-optimization techniques	Measure cost per successful task/prediction; select fit-for-purpose model/resources, bound context/steps, batch/cache safely, co-locate and retire idle artifacts.
Operational responsibilities	AI platform owns shared service patterns; domain product owns use case/data/quality; SRE owns reliability; Security/Privacy/Responsible AI own assurance controls.
Integration points	BigQuery, Cloud Storage, Knowledge Catalog, IAM, KMS, Logging/Monitoring, CI/CD, Artifact Registry, domain APIs and applicable AI services.
Alternatives	MLflow, custom BigQuery metadata, third-party experiment tracking.
Selection criteria	Use with Vertex AI development/pipelines when managed lineage integration is valuable.
Best practices	Use approved templates, immutable versions, regional isolation, risk-tier evidence, automated tests, least privilege, SLOs, cost budgets and tested rollback.
Reference architecture	Evidence layer connecting data, code, pipeline execution, model and deployment. Integrated through the enterprise AI platform, trust/control plane and regional data products.
Sample configuration	Run name usecase-env-date-sha; parameters, dataset snapshot, metrics, artifacts and release decision linked.
Common pitfalls	Logging secrets, PII or raw data samples in parameters/metadata.
Anti-patterns	Unmanaged console-only production, embedded credentials, raw/unclassified data, hard-coded mutable model identifiers, missing evaluation or no rollback.
Production-readiness checklist	Current product/location/quota/security validation; approved use case and data; IAM/network/KMS/logging; tests/evaluation; SLO/alerts/runbook; capacity/cost; DR and rollback evidence.

28.6 Vertex AI Feature Store

Assessment field	Guidance
Service name	Vertex AI Feature Store
Purpose	Manage and serve reusable ML features for online prediction with offline governance integration.
Enterprise architecture role	Conditional feature-serving plane; BigQuery remains primary offline feature/data plane.
Business capabilities supported	AI product delivery, decision support, enterprise knowledge access, workflow automation, risk control and operational efficiency.
Technical capabilities	Feature organization/discovery and online serving patterns integrated with Vertex AI/BigQuery capabilities.
Benefits	Reusable features, lower-latency lookup and improved online/offline consistency.
Limitations	Additional operational state and cost; capabilities and serving patterns must match current product version.
Service quotas	Validate current per-project, per-region, request, token, job, storage and concurrency quotas; establish quota owners, alerts and increase lead time.
Regional availability considerations	Validate the exact capability/model/version in AMER, EMEA and APAC target locations; co-locate data, runtime and telemetry and prohibit unapproved cross-region movement.
High-availability characteristics	Managed serving characteristics; clients require timeouts/fallback and source-based reconstruction.
Disaster-recovery considerations	Rebuild online features from BigQuery/source history and transformation code; define warm-up/reconciliation.
Security model	Entity keys, feature access and service accounts restricted by domain/use case; sensitive features minimized.
IAM requirements	Use least-privilege predefined/custom roles, dedicated workload identities, service-agent review, separation of builder/approver/deployer and no service-account keys.
Network requirements	Regional co-location with endpoints and source data; validate private/perimeter support.
VPC Service Controls considerations	Validate current inclusion, ingress/egress methods, supported APIs and dependent services; test perimeter paths and document exceptions.
Encryption capabilities	Google-managed encryption by default; validate in-transit protection and data/artifact encryption for each sub-capability.
CMEK support	Validate current resource/model/location-specific CMEK support and key lifecycle; do not claim unsupported coverage.
Logging	Enable required Admin Activity/Data Access/service logs and application correlation; minimize/redact sensitive prompts, features, evidence and outputs.
Monitoring	Monitor availability, errors, latency, throughput, quality/safety signals, quota, data freshness and unit economics.
Alerting	Route severity by risk tier to product/SRE/security; alert on telemetry absence, authorization/safety violations, SLO burn and budget anomalies.
Cost drivers	Online serving/storage, sync/materialization, BigQuery processing and network.
Cost-optimization techniques	Measure cost per successful task/prediction; select fit-for-purpose model/resources, bound context/steps, batch/cache safely, co-locate and retire idle artifacts.
Operational responsibilities	AI platform owns shared service patterns; domain product owns use case/data/quality; SRE owns reliability; Security/Privacy/Responsible AI own assurance controls.
Integration points	BigQuery, Cloud Storage, Knowledge Catalog, IAM, KMS, Logging/Monitoring, CI/CD, Artifact Registry, domain APIs and applicable AI services.
Alternatives	Direct BigQuery/batch features; Bigtable/Spanner/Redis custom serving; third-party feature stores.
Selection criteria	Use for shared low-latency online features with point-in-time and consistency requirements.
Best practices	Use approved templates, immutable versions, regional isolation, risk-tier evidence, automated tests, least privilege, SLOs, cost budgets and tested rollback.
Reference architecture	Conditional feature-serving plane; BigQuery remains primary offline feature/data plane. Integrated through the enterprise AI platform, trust/control plane and regional data products.
Sample configuration	Domain feature group keyed by tokenized customer ID with freshness and online/offline reconciliation SLO.
Common pitfalls	Adopting for every model or using direct identifiers and features without temporal semantics.
Anti-patterns	Unmanaged console-only production, embedded credentials, raw/unclassified data, hard-coded mutable model identifiers, missing evaluation or no rollback.
Production-readiness checklist	Current product/location/quota/security validation; approved use case and data; IAM/network/KMS/logging; tests/evaluation; SLO/alerts/runbook; capacity/cost; DR and rollback evidence.

28.7 Vertex AI Model Monitoring

Assessment field	Guidance
Service name	Vertex AI Model Monitoring
Purpose	Observe deployed model inputs, predictions, skew and drift where supported.
Enterprise architecture role	Managed component of model-behavior monitoring, supplemented by domain quality and business outcomes.
Business capabilities supported	AI product delivery, decision support, enterprise knowledge access, workflow automation, risk control and operational efficiency.
Technical capabilities	Model/version dependent skew/drift and monitoring workflows; generative/model monitoring maturity varies.
Benefits	Integrated telemetry and alerts for selected Vertex AI deployments.
Limitations	Coverage differs by monitoring version/model/deployment and some capabilities may be preview.
Service quotas	Validate current per-project, per-region, request, token, job, storage and concurrency quotas; establish quota owners, alerts and increase lead time.
Regional availability considerations	Validate the exact capability/model/version in AMER, EMEA and APAC target locations; co-locate data, runtime and telemetry and prohibit unapproved cross-region movement.
High-availability characteristics	Monitoring failure must not silently disable critical controls; alert on telemetry gaps.
Disaster-recovery considerations	Monitoring configs are IaC/config artifacts; restore after endpoint recovery and preserve reference data.
Security model	Monitoring datasets/logs are classified and access-controlled; samples minimized.
IAM requirements	Use least-privilege predefined/custom roles, dedicated workload identities, service-agent review, separation of builder/approver/deployer and no service-account keys.
Network requirements	Monitoring data paths and locations align with endpoint/data residency.
VPC Service Controls considerations	Validate current inclusion, ingress/egress methods, supported APIs and dependent services; test perimeter paths and document exceptions.
Encryption capabilities	Google-managed encryption by default; validate in-transit protection and data/artifact encryption for each sub-capability.
CMEK support	Validate current resource/model/location-specific CMEK support and key lifecycle; do not claim unsupported coverage.
Logging	Enable required Admin Activity/Data Access/service logs and application correlation; minimize/redact sensitive prompts, features, evidence and outputs.
Monitoring	Monitor availability, errors, latency, throughput, quality/safety signals, quota, data freshness and unit economics.
Alerting	Route severity by risk tier to product/SRE/security; alert on telemetry absence, authorization/safety violations, SLO burn and budget anomalies.
Cost drivers	Monitoring jobs, analysis/storage/logs and sampled data processing.
Cost-optimization techniques	Measure cost per successful task/prediction; select fit-for-purpose model/resources, bound context/steps, batch/cache safely, co-locate and retire idle artifacts.
Operational responsibilities	AI platform owns shared service patterns; domain product owns use case/data/quality; SRE owns reliability; Security/Privacy/Responsible AI own assurance controls.
Integration points	BigQuery, Cloud Storage, Knowledge Catalog, IAM, KMS, Logging/Monitoring, CI/CD, Artifact Registry, domain APIs and applicable AI services.
Alternatives	Custom BigQuery/Dataflow monitoring, Evidently/WhyLabs/third-party platforms.
Selection criteria	Use when selected model/deployment is supported and managed signals meet obligations.
Best practices	Use approved templates, immutable versions, regional isolation, risk-tier evidence, automated tests, least privilege, SLOs, cost budgets and tested rollback.
Reference architecture	Managed component of model-behavior monitoring, supplemented by domain quality and business outcomes. Integrated through the enterprise AI platform, trust/control plane and regional data products.
Sample configuration	Reference window, feature thresholds, alert route and incident playbook linked to model alias.
Common pitfalls	Assuming service availability equals quality monitoring or relying on preview-only controls for regulation.
Anti-patterns	Unmanaged console-only production, embedded credentials, raw/unclassified data, hard-coded mutable model identifiers, missing evaluation or no rollback.
Production-readiness checklist	Current product/location/quota/security validation; approved use case and data; IAM/network/KMS/logging; tests/evaluation; SLO/alerts/runbook; capacity/cost; DR and rollback evidence.

28.8 RAG Engine

Assessment field	Guidance
Service name	RAG Engine
Purpose	Managed corpus ingestion, chunking/embedding and retrieval integration for generative applications.
Enterprise architecture role	Application-controlled managed RAG option for approved enterprise corpora.
Business capabilities supported	AI product delivery, decision support, enterprise knowledge access, workflow automation, risk control and operational efficiency.
Technical capabilities	Corpus/file management, transformations, embeddings, retrieval and supported vector backends.
Benefits	Reduces custom RAG plumbing and integrates with Gemini/agent platform.
Limitations	Backend, location, connector, quota, deletion and security-control support are product dependent.
Service quotas	Validate current per-project, per-region, request, token, job, storage and concurrency quotas; establish quota owners, alerts and increase lead time.
Regional availability considerations	Validate the exact capability/model/version in AMER, EMEA and APAC target locations; co-locate data, runtime and telemetry and prohibit unapproved cross-region movement.
High-availability characteristics	Managed control/data paths; applications maintain retries, fallback and prior corpus/index alias.
Disaster-recovery considerations	Rebuild corpus/index from source documents, metadata/ACL export and versioned ingestion configuration.
Security model	Preserve ACL metadata, restrict corpus identity, validate VPC SC/CMEK, classify embeddings and audit retrieval.
IAM requirements	Use least-privilege predefined/custom roles, dedicated workload identities, service-agent review, separation of builder/approver/deployer and no service-account keys.
Network requirements	Co-locate with data/model; validate private/perimeter and backend connectivity.
VPC Service Controls considerations	Validate current inclusion, ingress/egress methods, supported APIs and dependent services; test perimeter paths and document exceptions.
Encryption capabilities	Google-managed encryption by default; validate in-transit protection and data/artifact encryption for each sub-capability.
CMEK support	Validate current resource/model/location-specific CMEK support and key lifecycle; do not claim unsupported coverage.
Logging	Enable required Admin Activity/Data Access/service logs and application correlation; minimize/redact sensitive prompts, features, evidence and outputs.
Monitoring	Monitor availability, errors, latency, throughput, quality/safety signals, quota, data freshness and unit economics.
Alerting	Route severity by risk tier to product/SRE/security; alert on telemetry absence, authorization/safety violations, SLO burn and budget anomalies.
Cost drivers	Ingestion, embeddings, vector storage/search, generation and source/storage operations.
Cost-optimization techniques	Measure cost per successful task/prediction; select fit-for-purpose model/resources, bound context/steps, batch/cache safely, co-locate and retire idle artifacts.
Operational responsibilities	AI platform owns shared service patterns; domain product owns use case/data/quality; SRE owns reliability; Security/Privacy/Responsible AI own assurance controls.
Integration points	BigQuery, Cloud Storage, Knowledge Catalog, IAM, KMS, Logging/Monitoring, CI/CD, Artifact Registry, domain APIs and applicable AI services.
Alternatives	Agent Search; Vector Search direct; BigQuery vector search; custom/open-source RAG.
Selection criteria	Use when managed corpus APIs and supported backend/location satisfy retrieval/security needs.
Best practices	Use approved templates, immutable versions, regional isolation, risk-tier evidence, automated tests, least privilege, SLOs, cost budgets and tested rollback.
Reference architecture	Application-controlled managed RAG option for approved enterprise corpora. Integrated through the enterprise AI platform, trust/control plane and regional data products.
Sample configuration	Immutable candidate corpus built from approved GCS path, structure-aware chunks, embedding model and ACL metadata; alias switched after tests.
Common pitfalls	Indexing documents without source ACL/deletion semantics or assuming all backends are globally available.
Anti-patterns	Unmanaged console-only production, embedded credentials, raw/unclassified data, hard-coded mutable model identifiers, missing evaluation or no rollback.
Production-readiness checklist	Current product/location/quota/security validation; approved use case and data; IAM/network/KMS/logging; tests/evaluation; SLO/alerts/runbook; capacity/cost; DR and rollback evidence.

28.9 Agent Search

Assessment field	Guidance
Service name	Agent Search
Purpose	Managed enterprise search and RAG over approved content and data sources.
Enterprise architecture role	Search-oriented knowledge experience and retrieval service where managed connectors/ranking/access features fit.
Business capabilities supported	AI product delivery, decision support, enterprise knowledge access, workflow automation, risk control and operational efficiency.
Technical capabilities	Enterprise search, content ingestion/connectors, ranking, grounding/search experiences and agent integration.
Benefits	Accelerates high-quality search experiences with less custom indexing and relevance engineering.
Limitations	Connector, ACL, region, indexing, customization and service-boundary capabilities require validation.
Service quotas	Validate current per-project, per-region, request, token, job, storage and concurrency quotas; establish quota owners, alerts and increase lead time.
Regional availability considerations	Validate the exact capability/model/version in AMER, EMEA and APAC target locations; co-locate data, runtime and telemetry and prohibit unapproved cross-region movement.
High-availability characteristics	Managed service characteristics; application handles timeout, fallback and degraded search.
Disaster-recovery considerations	Retain source-of-record and connector/config inventory; rebuild/reconnect in approved recovery location.
Security model	Source ACL propagation, identity mapping, tenant isolation, DLP and audit are mandatory.
IAM requirements	Use least-privilege predefined/custom roles, dedicated workload identities, service-agent review, separation of builder/approver/deployer and no service-account keys.
Network requirements	Validate source connector paths, private access, service perimeter and location.
VPC Service Controls considerations	Validate current inclusion, ingress/egress methods, supported APIs and dependent services; test perimeter paths and document exceptions.
Encryption capabilities	Google-managed encryption by default; validate in-transit protection and data/artifact encryption for each sub-capability.
CMEK support	Validate current resource/model/location-specific CMEK support and key lifecycle; do not claim unsupported coverage.
Logging	Enable required Admin Activity/Data Access/service logs and application correlation; minimize/redact sensitive prompts, features, evidence and outputs.
Monitoring	Monitor availability, errors, latency, throughput, quality/safety signals, quota, data freshness and unit economics.
Alerting	Route severity by risk tier to product/SRE/security; alert on telemetry absence, authorization/safety violations, SLO burn and budget anomalies.
Cost drivers	Indexed data, search/query operations, connectors, generation/grounding and related storage.
Cost-optimization techniques	Measure cost per successful task/prediction; select fit-for-purpose model/resources, bound context/steps, batch/cache safely, co-locate and retire idle artifacts.
Operational responsibilities	AI platform owns shared service patterns; domain product owns use case/data/quality; SRE owns reliability; Security/Privacy/Responsible AI own assurance controls.
Integration points	BigQuery, Cloud Storage, Knowledge Catalog, IAM, KMS, Logging/Monitoring, CI/CD, Artifact Registry, domain APIs and applicable AI services.
Alternatives	RAG Engine; Vector Search; BigQuery search/vector; third-party enterprise search.
Selection criteria	Use for broad enterprise search when managed connector/access/relevance capabilities satisfy requirements.
Best practices	Use approved templates, immutable versions, regional isolation, risk-tier evidence, automated tests, least privilege, SLOs, cost budgets and tested rollback.
Reference architecture	Search-oriented knowledge experience and retrieval service where managed connectors/ranking/access features fit. Integrated through the enterprise AI platform, trust/control plane and regional data products.
Sample configuration	Regional policy-search app mapped to enterprise groups and source document ACLs with citation-enabled answers.
Common pitfalls	Assuming connector access synchronization is immediate or complete without negative entitlement tests.
Anti-patterns	Unmanaged console-only production, embedded credentials, raw/unclassified data, hard-coded mutable model identifiers, missing evaluation or no rollback.
Production-readiness checklist	Current product/location/quota/security validation; approved use case and data; IAM/network/KMS/logging; tests/evaluation; SLO/alerts/runbook; capacity/cost; DR and rollback evidence.

28.10 Vertex AI Vector Search / Vector Search 2.0

Assessment field	Guidance
Service name	Vertex AI Vector Search / Vector Search 2.0
Purpose	Managed vector indexing and low-latency similarity retrieval at enterprise scale.
Enterprise architecture role	Dedicated vector backend for custom retrieval, recommendations and semantic similarity workloads.
Business capabilities supported	AI product delivery, decision support, enterprise knowledge access, workflow automation, risk control and operational efficiency.
Technical capabilities	Vector indexes, nearest-neighbor search, filtering and managed serving; specific features depend on version.
Benefits	High-scale managed vector retrieval and integration with Google Cloud AI services.
Limitations	Index lifecycle, update/delete semantics, regional support and version-specific features require design.
Service quotas	Validate current per-project, per-region, request, token, job, storage and concurrency quotas; establish quota owners, alerts and increase lead time.
Regional availability considerations	Validate the exact capability/model/version in AMER, EMEA and APAC target locations; co-locate data, runtime and telemetry and prohibit unapproved cross-region movement.
High-availability characteristics	Managed serving topology; client retries, capacity and regional recovery are still required.
Disaster-recovery considerations	Rebuild index from governed embeddings/source; preserve configuration and approved prior index alias.
Security model	Restrict endpoint/index access, enforce tenant/security metadata filters and classify vectors.
IAM requirements	Use least-privilege predefined/custom roles, dedicated workload identities, service-agent review, separation of builder/approver/deployer and no service-account keys.
Network requirements	Validate private endpoint/PSC/VPC SC and co-location requirements for selected version.
VPC Service Controls considerations	Validate current inclusion, ingress/egress methods, supported APIs and dependent services; test perimeter paths and document exceptions.
Encryption capabilities	Google-managed encryption by default; validate in-transit protection and data/artifact encryption for each sub-capability.
CMEK support	Validate current resource/model/location-specific CMEK support and key lifecycle; do not claim unsupported coverage.
Logging	Enable required Admin Activity/Data Access/service logs and application correlation; minimize/redact sensitive prompts, features, evidence and outputs.
Monitoring	Monitor availability, errors, latency, throughput, quality/safety signals, quota, data freshness and unit economics.
Alerting	Route severity by risk tier to product/SRE/security; alert on telemetry absence, authorization/safety violations, SLO burn and budget anomalies.
Cost drivers	Index build/storage, deployed capacity/search queries, network and embedding generation.
Cost-optimization techniques	Measure cost per successful task/prediction; select fit-for-purpose model/resources, bound context/steps, batch/cache safely, co-locate and retire idle artifacts.
Operational responsibilities	AI platform owns shared service patterns; domain product owns use case/data/quality; SRE owns reliability; Security/Privacy/Responsible AI own assurance controls.
Integration points	BigQuery, Cloud Storage, Knowledge Catalog, IAM, KMS, Logging/Monitoring, CI/CD, Artifact Registry, domain APIs and applicable AI services.
Alternatives	RAG Engine backend, Agent Search, BigQuery vector search, AlloyDB/Spanner/vector databases, open source.
Selection criteria	Use when scale, latency, filtering and custom retrieval justify a dedicated managed vector service.
Best practices	Use approved templates, immutable versions, regional isolation, risk-tier evidence, automated tests, least privilege, SLOs, cost budgets and tested rollback.
Reference architecture	Dedicated vector backend for custom retrieval, recommendations and semantic similarity workloads. Integrated through the enterprise AI platform, trust/control plane and regional data products.
Sample configuration	Regional indexes partitioned by corpus/tenant class, metadata filtering and blue-green alias publication.
Common pitfalls	One global index containing mixed tenants/classifications or missing deletion/re-embedding strategy.
Anti-patterns	Unmanaged console-only production, embedded credentials, raw/unclassified data, hard-coded mutable model identifiers, missing evaluation or no rollback.
Production-readiness checklist	Current product/location/quota/security validation; approved use case and data; IAM/network/KMS/logging; tests/evaluation; SLO/alerts/runbook; capacity/cost; DR and rollback evidence.

28.11 Agent Development Kit and Agent Runtime

Assessment field	Guidance
Service name	Agent Development Kit and Agent Runtime
Purpose	Develop, evaluate and deploy governed agents and multi-agent applications.
Enterprise architecture role	Preferred managed agent framework/runtime where language, location, network and operational support fit.
Business capabilities supported	AI product delivery, decision support, enterprise knowledge access, workflow automation, risk control and operational efficiency.
Technical capabilities	Agent orchestration, tools, subagents, deployment/runtime, sessions/observability and platform integrations.
Benefits	Standard agent engineering model and managed operational path integrated with Gemini services.
Limitations	SDK/runtime capabilities evolve; managed runtime may not satisfy every portability/network/control need.
Service quotas	Validate current per-project, per-region, request, token, job, storage and concurrency quotas; establish quota owners, alerts and increase lead time.
Regional availability considerations	Validate the exact capability/model/version in AMER, EMEA and APAC target locations; co-locate data, runtime and telemetry and prohibit unapproved cross-region movement.
High-availability characteristics	Runtime characteristics are validated; agents require step/time limits, retries, durable state where needed and fallback.
Disaster-recovery considerations	Redeploy immutable agent/tool release to recovery runtime; restore approved state stores and disable unsafe tools.
Security model	Dedicated runtime identity, allowlisted tools, policy gateway, tenant context, human confirmation and trace audit.
IAM requirements	Use least-privilege predefined/custom roles, dedicated workload identities, service-agent review, separation of builder/approver/deployer and no service-account keys.
Network requirements	Private service/tool connectivity, controlled egress and VPC SC/PSC where supported.
VPC Service Controls considerations	Validate current inclusion, ingress/egress methods, supported APIs and dependent services; test perimeter paths and document exceptions.
Encryption capabilities	Google-managed encryption by default; validate in-transit protection and data/artifact encryption for each sub-capability.
CMEK support	Validate current resource/model/location-specific CMEK support and key lifecycle; do not claim unsupported coverage.
Logging	Enable required Admin Activity/Data Access/service logs and application correlation; minimize/redact sensitive prompts, features, evidence and outputs.
Monitoring	Monitor availability, errors, latency, throughput, quality/safety signals, quota, data freshness and unit economics.
Alerting	Route severity by risk tier to product/SRE/security; alert on telemetry absence, authorization/safety violations, SLO burn and budget anomalies.
Cost drivers	Model tokens, runtime resources, sessions/memory, tools/APIs, retrieval and observability.
Cost-optimization techniques	Measure cost per successful task/prediction; select fit-for-purpose model/resources, bound context/steps, batch/cache safely, co-locate and retire idle artifacts.
Operational responsibilities	AI platform owns shared service patterns; domain product owns use case/data/quality; SRE owns reliability; Security/Privacy/Responsible AI own assurance controls.
Integration points	BigQuery, Cloud Storage, Knowledge Catalog, IAM, KMS, Logging/Monitoring, CI/CD, Artifact Registry, domain APIs and applicable AI services.
Alternatives	Cloud Run/GKE application runtime with custom orchestration; other agent frameworks/platforms.
Selection criteria	Use when managed deployment/observability accelerates delivery without compromising required controls.
Best practices	Use approved templates, immutable versions, regional isolation, risk-tier evidence, automated tests, least privilege, SLOs, cost budgets and tested rollback.
Reference architecture	Preferred managed agent framework/runtime where language, location, network and operational support fit. Integrated through the enterprise AI platform, trust/control plane and regional data products.
Sample configuration	ADK orchestrator with read-only search/analytics agents, typed tool gateway and approval service for writes.
Common pitfalls	General-purpose tools, shared privileged identity, unbounded agent loops or evaluating only final answers.
Anti-patterns	Unmanaged console-only production, embedded credentials, raw/unclassified data, hard-coded mutable model identifiers, missing evaluation or no rollback.
Production-readiness checklist	Current product/location/quota/security validation; approved use case and data; IAM/network/KMS/logging; tests/evaluation; SLO/alerts/runbook; capacity/cost; DR and rollback evidence.

28.12 Generative AI Evaluation Service

Assessment field	Guidance
Service name	Generative AI Evaluation Service
Purpose	Evaluate model, prompt, RAG and agent quality with managed metrics and judge-based methods.
Enterprise architecture role	Component of offline release gates and experiment comparison, calibrated with domain human evaluation.
Business capabilities supported	AI product delivery, decision support, enterprise knowledge access, workflow automation, risk control and operational efficiency.
Technical capabilities	Pointwise/pairwise metrics, custom rubrics, model-based evaluation and agent response/trajectory evaluation depending on current support.
Benefits	Standardized scalable evaluation integrated with platform models and development workflows.
Limitations	Judge bias, metric availability and preview capabilities; not a substitute for independent human validation.
Service quotas	Validate current per-project, per-region, request, token, job, storage and concurrency quotas; establish quota owners, alerts and increase lead time.
Regional availability considerations	Validate the exact capability/model/version in AMER, EMEA and APAC target locations; co-locate data, runtime and telemetry and prohibit unapproved cross-region movement.
High-availability characteristics	Evaluation is a release process; retry failed jobs and retain deterministic exported results.
Disaster-recovery considerations	Version datasets/rubrics/config and rerun in approved location/model if service unavailable.
Security model	Evaluation prompts/outputs may be sensitive; restrict datasets, minimize content and validate judge data use.
IAM requirements	Use least-privilege predefined/custom roles, dedicated workload identities, service-agent review, separation of builder/approver/deployer and no service-account keys.
Network requirements	Regional API and data co-location; validate perimeter/private paths.
VPC Service Controls considerations	Validate current inclusion, ingress/egress methods, supported APIs and dependent services; test perimeter paths and document exceptions.
Encryption capabilities	Google-managed encryption by default; validate in-transit protection and data/artifact encryption for each sub-capability.
CMEK support	Validate current resource/model/location-specific CMEK support and key lifecycle; do not claim unsupported coverage.
Logging	Enable required Admin Activity/Data Access/service logs and application correlation; minimize/redact sensitive prompts, features, evidence and outputs.
Monitoring	Monitor availability, errors, latency, throughput, quality/safety signals, quota, data freshness and unit economics.
Alerting	Route severity by risk tier to product/SRE/security; alert on telemetry absence, authorization/safety violations, SLO burn and budget anomalies.
Cost drivers	Judge/model calls, evaluated tokens, dataset processing and storage.
Cost-optimization techniques	Measure cost per successful task/prediction; select fit-for-purpose model/resources, bound context/steps, batch/cache safely, co-locate and retire idle artifacts.
Operational responsibilities	AI platform owns shared service patterns; domain product owns use case/data/quality; SRE owns reliability; Security/Privacy/Responsible AI own assurance controls.
Integration points	BigQuery, Cloud Storage, Knowledge Catalog, IAM, KMS, Logging/Monitoring, CI/CD, Artifact Registry, domain APIs and applicable AI services.
Alternatives	Custom evaluation harnesses, open-source frameworks, human evaluation platforms.
Selection criteria	Use for supported metrics/agent evaluation when calibrated and combined with deterministic/security tests.
Best practices	Use approved templates, immutable versions, regional isolation, risk-tier evidence, automated tests, least privilege, SLOs, cost budgets and tested rollback.
Reference architecture	Component of offline release gates and experiment comparison, calibrated with domain human evaluation. Integrated through the enterprise AI platform, trust/control plane and regional data products.
Sample configuration	Release job compares candidate and champion on task success, groundedness, citation and safety with a deterministic threshold aggregator.
Common pitfalls	Changing judge/rubric to make a candidate pass or using one aggregate score to hide critical failures.
Anti-patterns	Unmanaged console-only production, embedded credentials, raw/unclassified data, hard-coded mutable model identifiers, missing evaluation or no rollback.
Production-readiness checklist	Current product/location/quota/security validation; approved use case and data; IAM/network/KMS/logging; tests/evaluation; SLO/alerts/runbook; capacity/cost; DR and rollback evidence.

28.13 Document AI

Assessment field	Guidance
Service name	Document AI
Purpose	Extract, classify and structure information from enterprise documents.
Enterprise architecture role	Document-ingestion capability feeding governed products, RAG corpora and workflows.
Business capabilities supported	AI product delivery, decision support, enterprise knowledge access, workflow automation, risk control and operational efficiency.
Technical capabilities	OCR, layout, forms, specialized processors, custom extraction/classification and human review integrations depending on processor.
Benefits	Higher-quality structured extraction than generic OCR for supported document types.
Limitations	Processor availability, language/location, quotas, version behavior and confidence vary; human review is often required.
Service quotas	Validate current per-project, per-region, request, token, job, storage and concurrency quotas; establish quota owners, alerts and increase lead time.
Regional availability considerations	Validate the exact capability/model/version in AMER, EMEA and APAC target locations; co-locate data, runtime and telemetry and prohibit unapproved cross-region movement.
High-availability characteristics	Managed processor endpoint; ingestion pipeline provides retries, dead-letter, idempotency and manual fallback.
Disaster-recovery considerations	Preserve source documents and processor/version/config; reprocess in recovery location where supported.
Security model	Sensitive documents, CMEK/VPC SC/location support, retention and processor access validated; outputs inherit classification.
IAM requirements	Use least-privilege predefined/custom roles, dedicated workload identities, service-agent review, separation of builder/approver/deployer and no service-account keys.
Network requirements	Use approved regional endpoints and private/perimeter controls where supported.
VPC Service Controls considerations	Validate current inclusion, ingress/egress methods, supported APIs and dependent services; test perimeter paths and document exceptions.
Encryption capabilities	Google-managed encryption by default; validate in-transit protection and data/artifact encryption for each sub-capability.
CMEK support	Validate current resource/model/location-specific CMEK support and key lifecycle; do not claim unsupported coverage.
Logging	Enable required Admin Activity/Data Access/service logs and application correlation; minimize/redact sensitive prompts, features, evidence and outputs.
Monitoring	Monitor availability, errors, latency, throughput, quality/safety signals, quota, data freshness and unit economics.
Alerting	Route severity by risk tier to product/SRE/security; alert on telemetry absence, authorization/safety violations, SLO burn and budget anomalies.
Cost drivers	Pages processed, processor type, custom training/evaluation, storage and downstream processing.
Cost-optimization techniques	Measure cost per successful task/prediction; select fit-for-purpose model/resources, bound context/steps, batch/cache safely, co-locate and retire idle artifacts.
Operational responsibilities	AI platform owns shared service patterns; domain product owns use case/data/quality; SRE owns reliability; Security/Privacy/Responsible AI own assurance controls.
Integration points	BigQuery, Cloud Storage, Knowledge Catalog, IAM, KMS, Logging/Monitoring, CI/CD, Artifact Registry, domain APIs and applicable AI services.
Alternatives	Vision OCR, open-source/commercial OCR/extraction, manual processing.
Selection criteria	Use when document structure/extraction quality and managed processing justify it.
Best practices	Use approved templates, immutable versions, regional isolation, risk-tier evidence, automated tests, least privilege, SLOs, cost budgets and tested rollback.
Reference architecture	Document-ingestion capability feeding governed products, RAG corpora and workflows. Integrated through the enterprise AI platform, trust/control plane and regional data products.
Sample configuration	Claims form processor with confidence threshold, field validation, human exception queue and lineage to source page.
Common pitfalls	Treating extracted text/fields as authoritative without confidence, reconciliation and source reference.
Anti-patterns	Unmanaged console-only production, embedded credentials, raw/unclassified data, hard-coded mutable model identifiers, missing evaluation or no rollback.
Production-readiness checklist	Current product/location/quota/security validation; approved use case and data; IAM/network/KMS/logging; tests/evaluation; SLO/alerts/runbook; capacity/cost; DR and rollback evidence.

28.14 Cloud Speech-to-Text

Assessment field	Guidance
Service name	Cloud Speech-to-Text
Purpose	Transcribe streaming or batch audio into text for approved enterprise workflows.
Enterprise architecture role	Specialized speech-ingestion service for contact center, accessibility and voice experiences.
Business capabilities supported	AI product delivery, decision support, enterprise knowledge access, workflow automation, risk control and operational efficiency.
Technical capabilities	Streaming/batch transcription, languages/models, diarization/adaptation and timestamps subject to current support.
Benefits	Managed scalable speech recognition integrated with Google Cloud.
Limitations	Accuracy varies by noise, accent, language and domain vocabulary; consent and regional availability matter.
Service quotas	Validate current per-project, per-region, request, token, job, storage and concurrency quotas; establish quota owners, alerts and increase lead time.
Regional availability considerations	Validate the exact capability/model/version in AMER, EMEA and APAC target locations; co-locate data, runtime and telemetry and prohibit unapproved cross-region movement.
High-availability characteristics	Retry batch, reconnect streaming, buffer/replay audio where lawful and provide manual fallback.
Disaster-recovery considerations	Preserve authorized source audio/config if retention permits; reprocess in recovery region/model.
Security model	Consent, encryption, minimization, redaction, retention, speaker/biometric considerations and restricted access.
IAM requirements	Use least-privilege predefined/custom roles, dedicated workload identities, service-agent review, separation of builder/approver/deployer and no service-account keys.
Network requirements	Approved regional endpoint and controlled streaming path.
VPC Service Controls considerations	Validate current inclusion, ingress/egress methods, supported APIs and dependent services; test perimeter paths and document exceptions.
Encryption capabilities	Google-managed encryption by default; validate in-transit protection and data/artifact encryption for each sub-capability.
CMEK support	Validate current resource/model/location-specific CMEK support and key lifecycle; do not claim unsupported coverage.
Logging	Enable required Admin Activity/Data Access/service logs and application correlation; minimize/redact sensitive prompts, features, evidence and outputs.
Monitoring	Monitor availability, errors, latency, throughput, quality/safety signals, quota, data freshness and unit economics.
Alerting	Route severity by risk tier to product/SRE/security; alert on telemetry absence, authorization/safety violations, SLO burn and budget anomalies.
Cost drivers	Audio duration, model/features, streaming/batch processing, storage and downstream AI.
Cost-optimization techniques	Measure cost per successful task/prediction; select fit-for-purpose model/resources, bound context/steps, batch/cache safely, co-locate and retire idle artifacts.
Operational responsibilities	AI platform owns shared service patterns; domain product owns use case/data/quality; SRE owns reliability; Security/Privacy/Responsible AI own assurance controls.
Integration points	BigQuery, Cloud Storage, Knowledge Catalog, IAM, KMS, Logging/Monitoring, CI/CD, Artifact Registry, domain APIs and applicable AI services.
Alternatives	Third-party/on-prem speech recognition; manual transcription.
Selection criteria	Use when supported language/model/location meets measured domain accuracy and compliance.
Best practices	Use approved templates, immutable versions, regional isolation, risk-tier evidence, automated tests, least privilege, SLOs, cost budgets and tested rollback.
Reference architecture	Specialized speech-ingestion service for contact center, accessibility and voice experiences. Integrated through the enterprise AI platform, trust/control plane and regional data products.
Sample configuration	Contact-center transcription with phrase adaptation, confidence flags, DLP redaction and sampled QA.
Common pitfalls	Using low-confidence transcripts as factual records or indexing audio/transcript without consent/retention controls.
Anti-patterns	Unmanaged console-only production, embedded credentials, raw/unclassified data, hard-coded mutable model identifiers, missing evaluation or no rollback.
Production-readiness checklist	Current product/location/quota/security validation; approved use case and data; IAM/network/KMS/logging; tests/evaluation; SLO/alerts/runbook; capacity/cost; DR and rollback evidence.

28.15 Cloud Text-to-Speech

Assessment field	Guidance
Service name	Cloud Text-to-Speech
Purpose	Synthesize approved text into audio for accessible and conversational experiences.
Enterprise architecture role	Specialized output service behind content, disclosure and voice-policy controls.
Business capabilities supported	AI product delivery, decision support, enterprise knowledge access, workflow automation, risk control and operational efficiency.
Technical capabilities	Multiple voices/languages and synthesis formats/features subject to current availability.
Benefits	Managed scalable speech synthesis and broad channel integration.
Limitations	Voice availability/location, pronunciation and synthetic-media rights/policies require validation.
Service quotas	Validate current per-project, per-region, request, token, job, storage and concurrency quotas; establish quota owners, alerts and increase lead time.
Regional availability considerations	Validate the exact capability/model/version in AMER, EMEA and APAC target locations; co-locate data, runtime and telemetry and prohibit unapproved cross-region movement.
High-availability characteristics	Retry/cached safe audio and fallback to text or alternate approved voice.
Disaster-recovery considerations	Regenerate from approved text/config or use pre-generated critical prompts.
Security model	Restrict voice cloning/custom voice, protect text input, disclose synthetic content and prevent impersonation.
IAM requirements	Use least-privilege predefined/custom roles, dedicated workload identities, service-agent review, separation of builder/approver/deployer and no service-account keys.
Network requirements	Approved endpoint and egress/channel path.
VPC Service Controls considerations	Validate current inclusion, ingress/egress methods, supported APIs and dependent services; test perimeter paths and document exceptions.
Encryption capabilities	Google-managed encryption by default; validate in-transit protection and data/artifact encryption for each sub-capability.
CMEK support	Validate current resource/model/location-specific CMEK support and key lifecycle; do not claim unsupported coverage.
Logging	Enable required Admin Activity/Data Access/service logs and application correlation; minimize/redact sensitive prompts, features, evidence and outputs.
Monitoring	Monitor availability, errors, latency, throughput, quality/safety signals, quota, data freshness and unit economics.
Alerting	Route severity by risk tier to product/SRE/security; alert on telemetry absence, authorization/safety violations, SLO burn and budget anomalies.
Cost drivers	Characters/audio generation, premium voices/features, storage and delivery.
Cost-optimization techniques	Measure cost per successful task/prediction; select fit-for-purpose model/resources, bound context/steps, batch/cache safely, co-locate and retire idle artifacts.
Operational responsibilities	AI platform owns shared service patterns; domain product owns use case/data/quality; SRE owns reliability; Security/Privacy/Responsible AI own assurance controls.
Integration points	BigQuery, Cloud Storage, Knowledge Catalog, IAM, KMS, Logging/Monitoring, CI/CD, Artifact Registry, domain APIs and applicable AI services.
Alternatives	Third-party speech synthesis; prerecorded human audio.
Selection criteria	Use for accessibility/voice UX when quality, rights, disclosure and location are approved.
Best practices	Use approved templates, immutable versions, regional isolation, risk-tier evidence, automated tests, least privilege, SLOs, cost budgets and tested rollback.
Reference architecture	Specialized output service behind content, disclosure and voice-policy controls. Integrated through the enterprise AI platform, trust/control plane and regional data products.
Sample configuration	Authenticated employee assistant speaks non-sensitive responses with approved voice and visible disclosure.
Common pitfalls	Generating public or identity-like voice content without legal/brand/security review.
Anti-patterns	Unmanaged console-only production, embedded credentials, raw/unclassified data, hard-coded mutable model identifiers, missing evaluation or no rollback.
Production-readiness checklist	Current product/location/quota/security validation; approved use case and data; IAM/network/KMS/logging; tests/evaluation; SLO/alerts/runbook; capacity/cost; DR and rollback evidence.

28.16 Cloud Vision / Vision AI

Assessment field	Guidance
Service name	Cloud Vision / Vision AI
Purpose	Analyze images for text, labels, objects and approved visual intelligence tasks.
Enterprise architecture role	Specialized visual ingestion/inference capability feeding governed workflows and multimodal applications.
Business capabilities supported	AI product delivery, decision support, enterprise knowledge access, workflow automation, risk control and operational efficiency.
Technical capabilities	OCR/detection/classification and related APIs or custom vision patterns depending on current product.
Benefits	Managed image analysis without operating bespoke vision infrastructure.
Limitations	False positives/negatives, bias, surveillance/biometric constraints and API feature/location differences.
Service quotas	Validate current per-project, per-region, request, token, job, storage and concurrency quotas; establish quota owners, alerts and increase lead time.
Regional availability considerations	Validate the exact capability/model/version in AMER, EMEA and APAC target locations; co-locate data, runtime and telemetry and prohibit unapproved cross-region movement.
High-availability characteristics	Retry with idempotent image ID; queue/manual review for low confidence or critical outcomes.
Disaster-recovery considerations	Preserve approved source image and configuration if retention permits; reprocess.
Security model	Classify images, strip unnecessary metadata, restrict biometric/surveillance use and protect outputs.
IAM requirements	Use least-privilege predefined/custom roles, dedicated workload identities, service-agent review, separation of builder/approver/deployer and no service-account keys.
Network requirements	Approved regional/service endpoint and storage path.
VPC Service Controls considerations	Validate current inclusion, ingress/egress methods, supported APIs and dependent services; test perimeter paths and document exceptions.
Encryption capabilities	Google-managed encryption by default; validate in-transit protection and data/artifact encryption for each sub-capability.
CMEK support	Validate current resource/model/location-specific CMEK support and key lifecycle; do not claim unsupported coverage.
Logging	Enable required Admin Activity/Data Access/service logs and application correlation; minimize/redact sensitive prompts, features, evidence and outputs.
Monitoring	Monitor availability, errors, latency, throughput, quality/safety signals, quota, data freshness and unit economics.
Alerting	Route severity by risk tier to product/SRE/security; alert on telemetry absence, authorization/safety violations, SLO burn and budget anomalies.
Cost drivers	Images/features processed, custom model compute, storage and downstream operations.
Cost-optimization techniques	Measure cost per successful task/prediction; select fit-for-purpose model/resources, bound context/steps, batch/cache safely, co-locate and retire idle artifacts.
Operational responsibilities	AI platform owns shared service patterns; domain product owns use case/data/quality; SRE owns reliability; Security/Privacy/Responsible AI own assurance controls.
Integration points	BigQuery, Cloud Storage, Knowledge Catalog, IAM, KMS, Logging/Monitoring, CI/CD, Artifact Registry, domain APIs and applicable AI services.
Alternatives	Gemini multimodal, Document AI, open-source/custom vision, third-party services.
Selection criteria	Use where specialized visual API meets quality, latency, location and governance requirements.
Best practices	Use approved templates, immutable versions, regional isolation, risk-tier evidence, automated tests, least privilege, SLOs, cost budgets and tested rollback.
Reference architecture	Specialized visual ingestion/inference capability feeding governed workflows and multimodal applications. Integrated through the enterprise AI platform, trust/control plane and regional data products.
Sample configuration	Manufacturing defect detection routes low-confidence cases to inspectors and records source/evidence.
Common pitfalls	Using generic labels as high-impact decisions without domain validation and human review.
Anti-patterns	Unmanaged console-only production, embedded credentials, raw/unclassified data, hard-coded mutable model identifiers, missing evaluation or no rollback.
Production-readiness checklist	Current product/location/quota/security validation; approved use case and data; IAM/network/KMS/logging; tests/evaluation; SLO/alerts/runbook; capacity/cost; DR and rollback evidence.

28.17 Sensitive Data Protection

Assessment field	Guidance
Service name	Sensitive Data Protection
Purpose	Discover, classify, inspect and de-identify sensitive data used by AI workflows.
Enterprise architecture role	Cross-cutting privacy control for data/corpora, prompts, traces, outputs and evaluation datasets.
Business capabilities supported	AI product delivery, decision support, enterprise knowledge access, workflow automation, risk control and operational efficiency.
Technical capabilities	Built-in/custom detectors, inspection, profiling and transformation/tokenization patterns.
Benefits	Scalable standardized sensitive-data discovery and de-identification integrated with Google Cloud.
Limitations	Detection is probabilistic/context dependent; images, languages and custom identifiers need validation.
Service quotas	Validate current per-project, per-region, request, token, job, storage and concurrency quotas; establish quota owners, alerts and increase lead time.
Regional availability considerations	Validate the exact capability/model/version in AMER, EMEA and APAC target locations; co-locate data, runtime and telemetry and prohibit unapproved cross-region movement.
High-availability characteristics	Design asynchronous jobs/retries and fail-closed gates for required scans.
Disaster-recovery considerations	Configurations/templates are IaC/config; reapply and rescan reconstructed assets.
Security model	Strict template access, KMS/tokenization controls, no broad result exposure and protected findings.
IAM requirements	Use least-privilege predefined/custom roles, dedicated workload identities, service-agent review, separation of builder/approver/deployer and no service-account keys.
Network requirements	Service perimeter/location/data path validated for source and output.
VPC Service Controls considerations	Validate current inclusion, ingress/egress methods, supported APIs and dependent services; test perimeter paths and document exceptions.
Encryption capabilities	Google-managed encryption by default; validate in-transit protection and data/artifact encryption for each sub-capability.
CMEK support	Validate current resource/model/location-specific CMEK support and key lifecycle; do not claim unsupported coverage.
Logging	Enable required Admin Activity/Data Access/service logs and application correlation; minimize/redact sensitive prompts, features, evidence and outputs.
Monitoring	Monitor availability, errors, latency, throughput, quality/safety signals, quota, data freshness and unit economics.
Alerting	Route severity by risk tier to product/SRE/security; alert on telemetry absence, authorization/safety violations, SLO burn and budget anomalies.
Cost drivers	Bytes/content inspected, profiling, transformation, storage and workflow frequency.
Cost-optimization techniques	Measure cost per successful task/prediction; select fit-for-purpose model/resources, bound context/steps, batch/cache safely, co-locate and retire idle artifacts.
Operational responsibilities	AI platform owns shared service patterns; domain product owns use case/data/quality; SRE owns reliability; Security/Privacy/Responsible AI own assurance controls.
Integration points	BigQuery, Cloud Storage, Knowledge Catalog, IAM, KMS, Logging/Monitoring, CI/CD, Artifact Registry, domain APIs and applicable AI services.
Alternatives	Native policy tags/masking, custom detectors, third-party DLP/tokenization.
Selection criteria	Use for broad discovery/de-identification where supported, augmented by business rules.
Best practices	Use approved templates, immutable versions, regional isolation, risk-tier evidence, automated tests, least privilege, SLOs, cost budgets and tested rollback.
Reference architecture	Cross-cutting privacy control for data/corpora, prompts, traces, outputs and evaluation datasets. Integrated through the enterprise AI platform, trust/control plane and regional data products.
Sample configuration	Pre-index inspection redacts direct identifiers and attaches sensitivity metadata before corpus publication.
Common pitfalls	Assuming a clean scan proves absence of sensitive data or logging full findings broadly.
Anti-patterns	Unmanaged console-only production, embedded credentials, raw/unclassified data, hard-coded mutable model identifiers, missing evaluation or no rollback.
Production-readiness checklist	Current product/location/quota/security validation; approved use case and data; IAM/network/KMS/logging; tests/evaluation; SLO/alerts/runbook; capacity/cost; DR and rollback evidence.

28.18 BigQuery ML

Assessment field	Guidance
Service name	BigQuery ML
Purpose	Develop and run selected ML and embedding/vector workflows close to governed BigQuery data using SQL.
Enterprise architecture role	Accessible predictive/AI option for analytics teams and batch-centric use cases within the inherited data plane.
Business capabilities supported	AI product delivery, decision support, enterprise knowledge access, workflow automation, risk control and operational efficiency.
Technical capabilities	SQL-based model training/evaluation/prediction, remote model integrations and vector/embedding functions subject to current support.
Benefits	Reduces data movement, uses familiar SQL and integrates with BigQuery governance/capacity.
Limitations	Not every algorithm, tuning, online-serving or custom environment need is supported; remote models add dependencies.
Service quotas	Validate current per-project, per-region, request, token, job, storage and concurrency quotas; establish quota owners, alerts and increase lead time.
Regional availability considerations	Validate the exact capability/model/version in AMER, EMEA and APAC target locations; co-locate data, runtime and telemetry and prohibit unapproved cross-region movement.
High-availability characteristics	BigQuery managed execution; jobs are idempotent/retriable and outputs are governed tables.
Disaster-recovery considerations	Recreate model from SQL, source snapshot and parameters; preserve approved artifacts/exports where needed.
Security model	BigQuery IAM, row/column controls, reservations, authorized data and remote connection identities.
IAM requirements	Use least-privilege predefined/custom roles, dedicated workload identities, service-agent review, separation of builder/approver/deployer and no service-account keys.
Network requirements	Regional co-location and remote connection/data egress validated.
VPC Service Controls considerations	Validate current inclusion, ingress/egress methods, supported APIs and dependent services; test perimeter paths and document exceptions.
Encryption capabilities	Google-managed encryption by default; validate in-transit protection and data/artifact encryption for each sub-capability.
CMEK support	Validate current resource/model/location-specific CMEK support and key lifecycle; do not claim unsupported coverage.
Logging	Enable required Admin Activity/Data Access/service logs and application correlation; minimize/redact sensitive prompts, features, evidence and outputs.
Monitoring	Monitor availability, errors, latency, throughput, quality/safety signals, quota, data freshness and unit economics.
Alerting	Route severity by risk tier to product/SRE/security; alert on telemetry absence, authorization/safety violations, SLO burn and budget anomalies.
Cost drivers	Query bytes/slots, model training/prediction, remote model tokens and storage.
Cost-optimization techniques	Measure cost per successful task/prediction; select fit-for-purpose model/resources, bound context/steps, batch/cache safely, co-locate and retire idle artifacts.
Operational responsibilities	AI platform owns shared service patterns; domain product owns use case/data/quality; SRE owns reliability; Security/Privacy/Responsible AI own assurance controls.
Integration points	BigQuery, Cloud Storage, Knowledge Catalog, IAM, KMS, Logging/Monitoring, CI/CD, Artifact Registry, domain APIs and applicable AI services.
Alternatives	Vertex AI custom/AutoML; Spark ML; external platforms.
Selection criteria	Use for SQL-centric, batch and data-local workloads that meet algorithm/operational requirements.
Best practices	Use approved templates, immutable versions, regional isolation, risk-tier evidence, automated tests, least privilege, SLOs, cost budgets and tested rollback.
Reference architecture	Accessible predictive/AI option for analytics teams and batch-centric use cases within the inherited data plane. Integrated through the enterprise AI platform, trust/control plane and regional data products.
Sample configuration	Versioned SQL trains a forecasting/classification model from certified data with assertions and writes predictions as a data product.
Common pitfalls	Allowing uncontrolled analysts to publish production models or ignoring slot/query and remote-model costs.
Anti-patterns	Unmanaged console-only production, embedded credentials, raw/unclassified data, hard-coded mutable model identifiers, missing evaluation or no rollback.
Production-readiness checklist	Current product/location/quota/security validation; approved use case and data; IAM/network/KMS/logging; tests/evaluation; SLO/alerts/runbook; capacity/cost; DR and rollback evidence.

29. Architecture Decision Records

The following ADRs preserve inherited platform decisions and define Volume 07 choices. They use the full enterprise ADR structure and are reviewed after material product, model, scale, regulatory, security or commercial change.

ADR-AI-001 - Adopt Gemini Enterprise Agent Platform / Vertex AI as the primary managed AI platform

Field	Decision record
ADR ID	ADR-AI-001
Decision title	Adopt Gemini Enterprise Agent Platform / Vertex AI as the primary managed AI platform
Status	Approved
Date	11 July 2026
Architecture domain	AI platform
Problem	The enterprise needs a consistent, secure and operable platform for predictive ML, GenAI, RAG and agents across regions and domains.
Context	Global Fortune 100 AI platform operating across AMER, EMEA and APAC with regulated data, federated domains and inherited Google Cloud data/security foundations.
Decision drivers	Managed operations; Google Cloud data/security integration; MLOps and agent capabilities; Regional governance; Scalability
Constraints	Data residency, regulated/high-impact use cases, evolving managed services, petabyte-scale data, many domains and the need for reversible delivery.
Assumptions	Landing-zone, identity, data products, CI/CD, monitoring and governance foundations from prior volumes are available.
Alternatives considered	Separate point solutions only; Self-managed AI stack on GKE; Third-party primary AI platform; Managed Google Cloud AI platform with exceptions
Evaluation criteria	Security, privacy, responsible AI, quality, scalability, latency, operability, cost, location, interoperability and reversibility.
Decision	Use the managed Google Cloud AI platform as the default. Permit third-party or self-managed components only through workload-specific ADR and equivalent controls.
Rationale	It aligns with the inherited Google Cloud data platform and reduces duplicated identity, network, audit and operations patterns.
Benefits	Faster delivery, common controls and reduced platform fragmentation.
Tradeoffs	Platform dependency and evolving product boundaries; maintain abstraction, exportable artifacts and approved alternatives.
Positive consequences	Faster delivery, common controls and reduced platform fragmentation.
Negative consequences	Platform dependency and evolving product boundaries; maintain abstraction, exportable artifacts and approved alternatives.
Risks	A required capability/control is unavailable in a mandated region.; Commercial or quality outcomes become unacceptable.
Mitigations	Quarterly service validation; Stable model/tool interfaces; Portable source, prompts, evaluation and containers.
Implementation implications	Golden templates, policy/evaluation automation, regional projects and operating responsibilities must be funded and delivered.
Cost implications	Measure current service/model variables and unit economics; the decision may increase control/evaluation cost while reducing duplicated platforms and production risk.
Security implications	Least privilege, private/perimeter controls where supported, authorized data, audit, supply-chain assurance and incident disable/rollback are mandatory.
Operational implications	Product owner, AI platform, SRE, security and governance must jointly operate SLOs, releases, incidents, quotas and lifecycle.
Review triggers	Material service/model/version change; new regulation or geography; critical incident; inability to meet SLO/economics; approved alternative materially outperforms.
Review date	Quarterly; next formal review October 2026

ADR-AI-002 - Use an approved model catalog and stable model aliases

Field	Decision record
ADR ID	ADR-AI-002
Decision title	Use an approved model catalog and stable model aliases
Status	Approved
Date	11 July 2026
Architecture domain	Model architecture
Problem	Direct use of arbitrary or mutable model identifiers creates inconsistent quality, safety, cost and audit behavior.
Context	Global Fortune 100 AI platform operating across AMER, EMEA and APAC with regulated data, federated domains and inherited Google Cloud data/security foundations.
Decision drivers	Governance; Reproducibility; Model choice; Rollback; Cost control
Constraints	Data residency, regulated/high-impact use cases, evolving managed services, petabyte-scale data, many domains and the need for reversible delivery.
Assumptions	Landing-zone, identity, data products, CI/CD, monitoring and governance foundations from prior volumes are available.
Alternatives considered	Allow any Model Garden model; Hard-code version in each app; Central gateway only; Approved catalog plus application aliases
Evaluation criteria	Security, privacy, responsible AI, quality, scalability, latency, operability, cost, location, interoperability and reversibility.
Decision	Maintain an approved catalog by risk/data class. Applications call stable aliases that resolve to an evaluated model/version/configuration.
Rationale	Models change independently of application code; aliases and evidence make change controlled and reversible.
Benefits	Consistent selection and rapid rollback.
Tradeoffs	Catalog operations and potential lag in adopting new models.
Positive consequences	Consistent selection and rapid rollback.
Negative consequences	Catalog operations and potential lag in adopting new models.
Risks	Alias misconfiguration changes behavior unexpectedly.; Teams bypass the catalog.
Mitigations	IaC/config review; Runtime deny policy and telemetry; Alias resolution captured in trace.
Implementation implications	Golden templates, policy/evaluation automation, regional projects and operating responsibilities must be funded and delivered.
Cost implications	Measure current service/model variables and unit economics; the decision may increase control/evaluation cost while reducing duplicated platforms and production risk.
Security implications	Least privilege, private/perimeter controls where supported, authorized data, audit, supply-chain assurance and incident disable/rollback are mandatory.
Operational implications	Product owner, AI platform, SRE, security and governance must jointly operate SLOs, releases, incidents, quotas and lifecycle.
Review triggers	Material service/model/version change; new regulation or geography; critical incident; inability to meet SLO/economics; approved alternative materially outperforms.
Review date	Quarterly; next formal review October 2026

ADR-AI-003 - Adopt permission-aware RAG with evidence citations

Field	Decision record
ADR ID	ADR-AI-003
Decision title	Adopt permission-aware RAG with evidence citations
Status	Approved
Date	11 July 2026
Architecture domain	RAG / knowledge architecture
Problem	Enterprise answers require current knowledge while preserving source authorization and auditability.
Context	Global Fortune 100 AI platform operating across AMER, EMEA and APAC with regulated data, federated domains and inherited Google Cloud data/security foundations.
Decision drivers	Data authorization; Grounding; Freshness; Citations; Deletion and records controls
Constraints	Data residency, regulated/high-impact use cases, evolving managed services, petabyte-scale data, many domains and the need for reversible delivery.
Assumptions	Landing-zone, identity, data products, CI/CD, monitoring and governance foundations from prior volumes are available.
Alternatives considered	Prompt-only model knowledge; Fine-tuning on documents; Unfiltered vector index; Permission-aware RAG
Evaluation criteria	Security, privacy, responsible AI, quality, scalability, latency, operability, cost, location, interoperability and reversibility.
Decision	Use RAG where changing enterprise knowledge is required. Preserve source ACLs/metadata, filter before generation, cite material claims and abstain when evidence is insufficient.
Rationale	This provides updateability and evidence without granting the model broader access than the caller.
Benefits	More trustworthy and auditable answers.
Tradeoffs	Retrieval/index complexity and latency; use benchmarks, caching and managed options.
Positive consequences	More trustworthy and auditable answers.
Negative consequences	Retrieval/index complexity and latency; use benchmarks, caching and managed options.
Risks	ACL synchronization or deletion fails.; Citations are present but unsupported.
Mitigations	Negative entitlement tests; Deletion SLO; Claim-citation validation; Prior index rollback.
Implementation implications	Golden templates, policy/evaluation automation, regional projects and operating responsibilities must be funded and delivered.
Cost implications	Measure current service/model variables and unit economics; the decision may increase control/evaluation cost while reducing duplicated platforms and production risk.
Security implications	Least privilege, private/perimeter controls where supported, authorized data, audit, supply-chain assurance and incident disable/rollback are mandatory.
Operational implications	Product owner, AI platform, SRE, security and governance must jointly operate SLOs, releases, incidents, quotas and lifecycle.
Review triggers	Material service/model/version change; new regulation or geography; critical incident; inability to meet SLO/economics; approved alternative materially outperforms.
Review date	Quarterly; next formal review October 2026

ADR-AI-004 - Select vector and search backends per workload

Field	Decision record
ADR ID	ADR-AI-004
Decision title	Select vector and search backends per workload
Status	Approved
Date	11 July 2026
Architecture domain	Retrieval infrastructure
Problem	A single vector backend cannot optimize connector support, latency, hybrid search, filtering, governance, scale and cost for every use case.
Context	Global Fortune 100 AI platform operating across AMER, EMEA and APAC with regulated data, federated domains and inherited Google Cloud data/security foundations.
Decision drivers	Workload fit; Security/location; Scale/latency; Hybrid retrieval; Update/delete; Operations
Constraints	Data residency, regulated/high-impact use cases, evolving managed services, petabyte-scale data, many domains and the need for reversible delivery.
Assumptions	Landing-zone, identity, data products, CI/CD, monitoring and governance foundations from prior volumes are available.
Alternatives considered	One enterprise Vector Search cluster; RAG Engine only; Agent Search only; Workload decision tree across managed options
Evaluation criteria	Security, privacy, responsible AI, quality, scalability, latency, operability, cost, location, interoperability and reversibility.
Decision	Choose Agent Search, RAG Engine, Vector Search or BigQuery vector capabilities using documented selection criteria; standardize interfaces and evidence, not one backend.
Rationale	Managed services have different strengths and rapidly evolving boundaries.
Benefits	Better fit and avoids forced architecture.
Tradeoffs	More platform patterns to govern and support.
Positive consequences	Better fit and avoids forced architecture.
Negative consequences	More platform patterns to govern and support.
Risks	Fragmentation and duplicate indexes.; Unsupported regional feature selected.
Mitigations	Approved patterns; Architecture review; Shared evaluation/metadata contract; Current product validation.
Implementation implications	Golden templates, policy/evaluation automation, regional projects and operating responsibilities must be funded and delivered.
Cost implications	Measure current service/model variables and unit economics; the decision may increase control/evaluation cost while reducing duplicated platforms and production risk.
Security implications	Least privilege, private/perimeter controls where supported, authorized data, audit, supply-chain assurance and incident disable/rollback are mandatory.
Operational implications	Product owner, AI platform, SRE, security and governance must jointly operate SLOs, releases, incidents, quotas and lifecycle.
Review triggers	Material service/model/version change; new regulation or geography; critical incident; inability to meet SLO/economics; approved alternative materially outperforms.
Review date	Quarterly; next formal review October 2026

ADR-AI-005 - Use ADK and managed Agent Runtime as the preferred agent pattern

Field	Decision record
ADR ID	ADR-AI-005
Decision title	Use ADK and managed Agent Runtime as the preferred agent pattern
Status	Approved with workload validation
Date	11 July 2026
Architecture domain	Agent architecture
Problem	Agent teams need a standard development and deployment path with tools, observability and operational controls.
Context	Global Fortune 100 AI platform operating across AMER, EMEA and APAC with regulated data, federated domains and inherited Google Cloud data/security foundations.
Decision drivers	Developer productivity; Managed runtime; Tool governance; Observability; Portability
Constraints	Data residency, regulated/high-impact use cases, evolving managed services, petabyte-scale data, many domains and the need for reversible delivery.
Assumptions	Landing-zone, identity, data products, CI/CD, monitoring and governance foundations from prior volumes are available.
Alternatives considered	Custom agent framework on Cloud Run; Third-party agent runtime; ADK plus Agent Runtime; No agents
Evaluation criteria	Security, privacy, responsible AI, quality, scalability, latency, operability, cost, location, interoperability and reversibility.
Decision	Use ADK for new governed agents and Agent Runtime where current region/network/control support meets requirements. Use Cloud Run/custom runtime as an approved alternative.
Rationale	A standard framework reduces bespoke orchestration while retaining an exit path.
Benefits	Reusable agent engineering and operations.
Tradeoffs	SDK/runtime evolution and possible gaps; pin versions and maintain portable agent/tool contracts.
Positive consequences	Reusable agent engineering and operations.
Negative consequences	SDK/runtime evolution and possible gaps; pin versions and maintain portable agent/tool contracts.
Risks	Runtime lacks required private/regional feature.; Agent behavior or cost is unstable.
Mitigations	Workload proof; Version pinning; Canary; Step/tool budgets; Cloud Run fallback.
Implementation implications	Golden templates, policy/evaluation automation, regional projects and operating responsibilities must be funded and delivered.
Cost implications	Measure current service/model variables and unit economics; the decision may increase control/evaluation cost while reducing duplicated platforms and production risk.
Security implications	Least privilege, private/perimeter controls where supported, authorized data, audit, supply-chain assurance and incident disable/rollback are mandatory.
Operational implications	Product owner, AI platform, SRE, security and governance must jointly operate SLOs, releases, incidents, quotas and lifecycle.
Review triggers	Material service/model/version change; new regulation or geography; critical incident; inability to meet SLO/economics; approved alternative materially outperforms.
Review date	Quarterly; next formal review October 2026

ADR-AI-006 - Maintain distinct but integrated MLOps and LLMOps controls

Field	Decision record
ADR ID	ADR-AI-006
Decision title	Maintain distinct but integrated MLOps and LLMOps controls
Status	Approved
Date	11 July 2026
Architecture domain	Lifecycle architecture
Problem	Classic model lifecycle controls do not fully cover prompts, corpora, retrieval, tools and emergent agent behavior.
Context	Global Fortune 100 AI platform operating across AMER, EMEA and APAC with regulated data, federated domains and inherited Google Cloud data/security foundations.
Decision drivers	Complete release manifest; Evaluation breadth; Drift; Audit; Operational ownership
Constraints	Data residency, regulated/high-impact use cases, evolving managed services, petabyte-scale data, many domains and the need for reversible delivery.
Assumptions	Landing-zone, identity, data products, CI/CD, monitoring and governance foundations from prior volumes are available.
Alternatives considered	Use MLOps unchanged; Separate unrelated GenAI process; Integrated lifecycle with LLMOps extensions
Evaluation criteria	Security, privacy, responsible AI, quality, scalability, latency, operability, cost, location, interoperability and reversibility.
Decision	Reuse pipeline, registry, metadata, CI/CD and SRE foundations while adding prompt, index, tool, trajectory, grounding and safety controls.
Rationale	This avoids duplicate platforms yet recognizes GenAI-specific failure modes.
Benefits	Consistent governance and reusable delivery.
Tradeoffs	More artifacts and tests per release.
Positive consequences	Consistent governance and reusable delivery.
Negative consequences	More artifacts and tests per release.
Risks	Teams register only model but not prompt/index/tool state.; Evaluation misses emergent behavior.
Mitigations	Release manifest schema; Risk-tier test suites; Trace correlation; Independent validation.
Implementation implications	Golden templates, policy/evaluation automation, regional projects and operating responsibilities must be funded and delivered.
Cost implications	Measure current service/model variables and unit economics; the decision may increase control/evaluation cost while reducing duplicated platforms and production risk.
Security implications	Least privilege, private/perimeter controls where supported, authorized data, audit, supply-chain assurance and incident disable/rollback are mandatory.
Operational implications	Product owner, AI platform, SRE, security and governance must jointly operate SLOs, releases, incidents, quotas and lifecycle.
Review triggers	Material service/model/version change; new regulation or geography; critical incident; inability to meet SLO/economics; approved alternative materially outperforms.
Review date	Quarterly; next formal review October 2026

ADR-AI-007 - Use Vertex AI Feature Store only for justified online feature use cases

Field	Decision record
ADR ID	ADR-AI-007
Decision title	Use Vertex AI Feature Store only for justified online feature use cases
Status	Approved
Date	11 July 2026
Architecture domain	Feature architecture
Problem	Universal feature-store adoption can add state and cost without benefit for batch-centric models.
Context	Global Fortune 100 AI platform operating across AMER, EMEA and APAC with regulated data, federated domains and inherited Google Cloud data/security foundations.
Decision drivers	Latency; Reuse; Point-in-time correctness; Online/offline consistency; Operations
Constraints	Data residency, regulated/high-impact use cases, evolving managed services, petabyte-scale data, many domains and the need for reversible delivery.
Assumptions	Landing-zone, identity, data products, CI/CD, monitoring and governance foundations from prior volumes are available.
Alternatives considered	Feature Store for all features; BigQuery only; Custom serving store; Conditional Feature Store
Evaluation criteria	Security, privacy, responsible AI, quality, scalability, latency, operability, cost, location, interoperability and reversibility.
Decision	Use BigQuery as the offline feature plane. Add Feature Store when reusable low-latency online features and consistency requirements justify it.
Rationale	This preserves the inherited data platform and avoids unnecessary complexity.
Benefits	Focused investment and lower platform sprawl.
Tradeoffs	Multiple feature delivery patterns; enforce a documented decision tree.
Positive consequences	Focused investment and lower platform sprawl.
Negative consequences	Multiple feature delivery patterns; enforce a documented decision tree.
Risks	Teams build bespoke online stores.; Feature skew between paths.
Mitigations	Platform pattern; Reconciliation tests; Ownership/SLO; Architecture review.
Implementation implications	Golden templates, policy/evaluation automation, regional projects and operating responsibilities must be funded and delivered.
Cost implications	Measure current service/model variables and unit economics; the decision may increase control/evaluation cost while reducing duplicated platforms and production risk.
Security implications	Least privilege, private/perimeter controls where supported, authorized data, audit, supply-chain assurance and incident disable/rollback are mandatory.
Operational implications	Product owner, AI platform, SRE, security and governance must jointly operate SLOs, releases, incidents, quotas and lifecycle.
Review triggers	Material service/model/version change; new regulation or geography; critical incident; inability to meet SLO/economics; approved alternative materially outperforms.
Review date	Quarterly; next formal review October 2026

ADR-AI-008 - Require human confirmation for high-impact or irreversible agent actions

Field	Decision record
ADR ID	ADR-AI-008
Decision title	Require human confirmation for high-impact or irreversible agent actions
Status	Approved
Date	11 July 2026
Architecture domain	Responsible AI / security
Problem	Probabilistic agent planning and tool use can create financial, legal, safety or access consequences.
Context	Global Fortune 100 AI platform operating across AMER, EMEA and APAC with regulated data, federated domains and inherited Google Cloud data/security foundations.
Decision drivers	Human agency; Authorization; Auditability; Reversibility; Risk proportionality
Constraints	Data residency, regulated/high-impact use cases, evolving managed services, petabyte-scale data, many domains and the need for reversible delivery.
Assumptions	Landing-zone, identity, data products, CI/CD, monitoring and governance foundations from prior volumes are available.
Alternatives considered	Fully autonomous actions; Manual workflows only; Human confirmation for high-impact actions
Evaluation criteria	Security, privacy, responsible AI, quality, scalability, latency, operability, cost, location, interoperability and reversibility.
Decision	Agents may prepare and recommend, but high-impact, irreversible, externally visible or privilege-changing actions require an authenticated human confirmation unless a specific autonomous policy is approved.
Rationale	A model cannot be the sole authority for consequential enterprise actions by default.
Benefits	Reduces harm and unauthorized action.
Tradeoffs	Additional latency and reviewer workload.
Positive consequences	Reduces harm and unauthorized action.
Negative consequences	Additional latency and reviewer workload.
Risks	Rubber-stamp review.; Approval context is incomplete or spoofed.
Mitigations	Clear action preview/evidence; Independent policy check; Reviewer competence; Immutable approval record.
Implementation implications	Golden templates, policy/evaluation automation, regional projects and operating responsibilities must be funded and delivered.
Cost implications	Measure current service/model variables and unit economics; the decision may increase control/evaluation cost while reducing duplicated platforms and production risk.
Security implications	Least privilege, private/perimeter controls where supported, authorized data, audit, supply-chain assurance and incident disable/rollback are mandatory.
Operational implications	Product owner, AI platform, SRE, security and governance must jointly operate SLOs, releases, incidents, quotas and lifecycle.
Review triggers	Material service/model/version change; new regulation or geography; critical incident; inability to meet SLO/economics; approved alternative materially outperforms.
Review date	Quarterly; next formal review October 2026

ADR-AI-009 - Deploy regional AI planes aligned with data residency

Field	Decision record
ADR ID	ADR-AI-009
Decision title	Deploy regional AI planes aligned with data residency
Status	Approved
Date	11 July 2026
Architecture domain	Region and resilience
Problem	Centralizing AI in one geography can violate residency, create latency/egress and hinder recovery.
Context	Global Fortune 100 AI platform operating across AMER, EMEA and APAC with regulated data, federated domains and inherited Google Cloud data/security foundations.
Decision drivers	Residency; Latency; Data sovereignty; Failure isolation; Cost
Constraints	Data residency, regulated/high-impact use cases, evolving managed services, petabyte-scale data, many domains and the need for reversible delivery.
Assumptions	Landing-zone, identity, data products, CI/CD, monitoring and governance foundations from prior volumes are available.
Alternatives considered	Single global AI project; Cross-region central endpoints; Regional AI planes with global policy
Evaluation criteria	Security, privacy, responsible AI, quality, scalability, latency, operability, cost, location, interoperability and reversibility.
Decision	Operate AMER, EMEA and APAC AI projects/corpora/endpoints aligned with source data. Global governance distributes policy and aggregates minimized telemetry.
Rationale	Regionalization preserves inherited data-plane boundaries and limits unnecessary movement.
Benefits	Improved compliance and latency.
Tradeoffs	Duplicated models/indexes and operational cost.
Positive consequences	Improved compliance and latency.
Negative consequences	Duplicated models/indexes and operational cost.
Risks	Capability unavailable in a region.; Cross-region telemetry leaks sensitive content.
Mitigations	Capability matrix; Approved exceptions; Aggregate/redact telemetry; Recovery design.
Implementation implications	Golden templates, policy/evaluation automation, regional projects and operating responsibilities must be funded and delivered.
Cost implications	Measure current service/model variables and unit economics; the decision may increase control/evaluation cost while reducing duplicated platforms and production risk.
Security implications	Least privilege, private/perimeter controls where supported, authorized data, audit, supply-chain assurance and incident disable/rollback are mandatory.
Operational implications	Product owner, AI platform, SRE, security and governance must jointly operate SLOs, releases, incidents, quotas and lifecycle.
Review triggers	Material service/model/version change; new regulation or geography; critical incident; inability to meet SLO/economics; approved alternative materially outperforms.
Review date	Quarterly; next formal review October 2026

ADR-AI-010 - Make evaluation a mandatory release gate

Field	Decision record
ADR ID	ADR-AI-010
Decision title	Make evaluation a mandatory release gate
Status	Approved
Date	11 July 2026
Architecture domain	Quality and assurance
Problem	AI behavior can regress after changes to model, prompt, data, retrieval, safety or tools even when code tests pass.
Context	Global Fortune 100 AI platform operating across AMER, EMEA and APAC with regulated data, federated domains and inherited Google Cloud data/security foundations.
Decision drivers	Quality; Safety; Grounding; Bias; Performance; Cost; Audit
Constraints	Data residency, regulated/high-impact use cases, evolving managed services, petabyte-scale data, many domains and the need for reversible delivery.
Assumptions	Landing-zone, identity, data products, CI/CD, monitoring and governance foundations from prior volumes are available.
Alternatives considered	Manual demo approval; Post-production monitoring only; Automated and human-calibrated evaluation gates
Evaluation criteria	Security, privacy, responsible AI, quality, scalability, latency, operability, cost, location, interoperability and reversibility.
Decision	Every production release executes a risk-tier evaluation suite and blocks when critical thresholds or approvals fail. High-risk releases include independent validation and red-team evidence.
Rationale	Evaluation converts subjective demos into repeatable evidence and supports rollback comparison.
Benefits	Lower escaped defects and auditable promotion.
Tradeoffs	Dataset maintenance and judge/human evaluation cost.
Positive consequences	Lower escaped defects and auditable promotion.
Negative consequences	Dataset maintenance and judge/human evaluation cost.
Risks	Metric gaming or stale test set.; Model judge bias.
Mitigations	Independent dataset ownership; Periodic refresh; Human calibration; Critical-metric vetoes.
Implementation implications	Golden templates, policy/evaluation automation, regional projects and operating responsibilities must be funded and delivered.
Cost implications	Measure current service/model variables and unit economics; the decision may increase control/evaluation cost while reducing duplicated platforms and production risk.
Security implications	Least privilege, private/perimeter controls where supported, authorized data, audit, supply-chain assurance and incident disable/rollback are mandatory.
Operational implications	Product owner, AI platform, SRE, security and governance must jointly operate SLOs, releases, incidents, quotas and lifecycle.
Review triggers	Material service/model/version change; new regulation or geography; critical incident; inability to meet SLO/economics; approved alternative materially outperforms.
Review date	Quarterly; next formal review October 2026

30. Risks and Mitigations

ID	Risk	Probability	Impact	Mitigation / control owner
AI-R01	Use case lacks accountable business purpose or owner.	High	High	Intake gate and portfolio review; AI Governance Council.
AI-R02	Training or retrieval uses unauthorized, stale or poor-quality data.	High	Critical	Certified products/corpora, lineage, ACL tests, quality/freshness and deletion SLO; Data/Knowledge owner.
AI-R03	Model or prompt change causes silent behavior regression.	High	High	Immutable release manifest, champion comparison and mandatory evaluation; AI Product.
AI-R04	RAG exposes another user/tenant or revoked document.	Medium	Critical	Pre-generation entitlement filtering, negative tests, deletion propagation and cache isolation; Security/Knowledge owner.
AI-R05	Prompt injection causes policy override or unsafe tool use.	High	Critical	Instruction separation, narrow tools, policy gateway, egress control, red team and human confirmation; Security/Agent owner.
AI-R06	Agent executes duplicate or irreversible actions.	Medium	Critical	Idempotency, typed tools, action preview, approval, transaction reconciliation and emergency disable; Application owner.
AI-R07	Model performs inequitably for affected segments.	Medium	Critical	Representativeness review, fairness metrics, independent validation, monitoring and appeal; Responsible AI.
AI-R08	Sensitive prompts, features, evidence or outputs leak through logs.	Medium	Critical	Telemetry minimization/redaction, restricted log views, retention and DLP sampling; Security/SRE.
AI-R09	Preview or region-limited feature becomes a production dependency.	Medium	High	Current product validation, compensating control, portable fallback and review trigger; Architecture.
AI-R10	Endpoint/model/vector quotas cause service failure.	Medium	High	Capacity model, quota owner, alerts, increase lead time, load test and fallback; Platform/SRE.
AI-R11	Token, endpoint or agent-tool cost exceeds business value.	High	High	Unit economics, budgets, model routing, context/step limits and retirement; FinOps/Product.
AI-R12	Model provider/version retirement breaks application.	Medium	High	Approved alternatives, alias abstraction, regression suite and migration runway; AI Platform.
AI-R13	Corpus/index cannot be reconstructed after corruption.	Low	Critical	Source-of-record, config/metadata export, blue-green indexes and rebuild exercise; Knowledge Platform.
AI-R14	Human review becomes a rubber stamp.	Medium	High	Competency, sampled QA, workload/capacity, clear evidence and override monitoring; Business/Risk.
AI-R15	Third-party/open model lacks equivalent controls or terms.	Medium	High	Vendor/model due diligence, deployment ADR, legal/security review and same release gates; Procurement/Architecture.
AI-R16	Evaluation dataset becomes stale or optimized against.	High	High	Independent ownership, refresh cadence, hidden holdout/adversarial cases and production incident feedback; Validation.
AI-R17	Regional duplication creates inconsistent models/prompts/indexes.	Medium	High	Global release manifests, regional promotion evidence and drift comparison; Platform Release.
AI-R18	AI system is unavailable with no business fallback.	Medium	Critical	Tiered DR, deterministic/manual fallback, exercises and business continuity procedure; Product/SRE.

30.1 Anti-patterns

Anti-pattern	Why it fails	Required correction
Demo-to-production	No risk, data, evaluation, SLO or rollback evidence.	Re-enter through use-case and release governance.
One unrestricted enterprise index	Breaks tenant, classification, residency and lifecycle boundaries.	Partition by authorization/risk and enforce pre-generation filters.
Prompt as security boundary	Text cannot enforce IAM or stop an authorized tool from overreaching.	Server-side identity, policy, schemas and least-privilege tools.
One largest model for all tasks	Unnecessary cost/latency and unpredictable change.	Model routing/selection by task evidence.
Raw trace everything	Creates a sensitive secondary data lake.	Telemetry minimization, redaction, access and retention.
Autonomous write tools by default	Probabilistic plans can create irreversible harm.	Read-only default, policy gateway and human approval.
Fine-tune before retrieval/data quality	Encodes stale/unauthorized information and complicates deletion.	First improve governed context and evaluate RAG/base model.
Treat citations as automatically correct	A URL can be irrelevant, inaccessible or unsupported.	Claim-evidence validation and authorization.

31. Implementation Roadmap and Migration

Wave	Indicative duration	Objectives and deliverables	Exit criteria
0. Mobilize and govern	4-6 weeks	AI portfolio, risk tiers, prohibited-use policy, project/region plan, approved model policy, operating RACI and first use cases.	Council/owners established; use cases and risk classifications approved.
1. AI platform foundation	6-10 weeks	Regional AI projects, APIs, IAM, network/perimeters, KMS, logging, artifact/pipeline roots, CI/CD and developer templates.	Security architecture and nonprod control tests pass.
2. MLOps MVP	8-12 weeks	Pipelines, experiments/metadata, registry, batch/online deployment, model card, monitoring and one predictive use case.	Reproducible train-to-rollback and SLO proof.
3. GenAI/LLMOps MVP	8-12 weeks	Approved model aliases, prompt repository, evaluation harness, safety/red-team suite, tracing and one low/moderate-risk assistant.	Release gate and online monitoring operational.
4. Enterprise RAG	10-14 weeks	Source registration, Document AI/DLP as needed, permission-aware corpus/index, retrieval/reranking, citations, deletion and DR.	Authorization, quality, load and rebuild tests pass.
5. Governed agents	10-16 weeks	ADK/runtime pattern, tool gateway, action policy, approval, trajectory evaluation, emergency disable and one bounded workflow.	Zero unauthorized actions in tests; business/SRE/security acceptance.
6. Scale and optimize	Ongoing	Domain onboarding, model/backend portfolio, regional expansion, FinOps, advanced monitoring, resilience and retirement.	Repeatable onboarding and quarterly assurance.

31.1 Migration approach

Current pattern	Target transition
Isolated notebooks and scripts	Inventory; classify; convert valuable workloads into reusable pipeline components and retire orphan experiments.
Third-party/on-prem ML serving	Benchmark; preserve approved workloads where justified; migrate identity, audit, registry and monitoring in waves.
Unmanaged GenAI API use	Block/route through approved projects/models; move prompts and evaluation to repositories; remediate sensitive data.
Search/chatbots without ACL-aware RAG	Register sources, preserve entitlement metadata, rebuild corpus/index and perform cross-user negative testing.
Robotic/process automation with broad credentials	Wrap operations as narrow APIs/tools with workload identity, idempotency, policy and approval.
Duplicate domain copilots	Consolidate shared platform patterns while preserving domain-specific knowledge and ownership.

31.2 Measurable rollout outcomes

- 100% of production AI use cases registered, risk-tiered and owned.
- 100% of releases have immutable model/prompt/index/agent manifests and rollback evidence.
- 100% of Tier 2/3 releases execute approved quality, safety and security evaluation suites.
- Zero critical cross-tenant retrieval or unauthorized tool action in acceptance and ongoing security tests.
- All Tier 0/1 AI products have SLOs, alerts, runbooks, on-call and tested fallback/DR.
- Cost per successful task is reported by domain/use case and reviewed monthly.

32. Validation, Production Readiness and Appendices

32.1 Architecture validation criteria

ID	Validation / acceptance evidence
VAL-AI-01	Every production use case has owner, purpose, affected users, risk tier and approval evidence.
VAL-AI-02	Training/retrieval/evaluation inputs map to authorized products or registered sources with lineage and region.
VAL-AI-03	Human and workload identities enforce least privilege and separation of builder/approver/deployer.
VAL-AI-04	Network, VPC SC, PSC/private access, CMEK and location support are validated for each selected sub-capability.
VAL-AI-05	Model, prompt, corpus/index, agent, tool, evaluation and runtime configuration are independently versioned and resolved in the release manifest.
VAL-AI-06	Quality, grounding, citations, safety, security, bias, latency, concurrency and cost gates pass by risk tier.
VAL-AI-07	RAG denies unauthorized/expired/deleted evidence and caches/memory cannot cross tenant or identity.
VAL-AI-08	Agent tools reauthorize server-side, validate schema, use idempotency and require human confirmation where mandated.
VAL-AI-09	Tier 0/1 SLO and fallback behavior pass load, quota and dependency-failure tests.
VAL-AI-10	Rollback to prior model/prompt/index/agent/tool release is tested and within RTO.
VAL-AI-11	Corpus/index and model deployment can be reconstructed from authoritative artifacts.
VAL-AI-12	Telemetry is sufficient for audit and incident response but minimizes sensitive content.
VAL-AI-13	Unit economics are within approved envelope at forecast and stress volume.
VAL-AI-14	Current official product names, SDKs, quotas, regions, security controls and commercial terms are signed off.

32.2 Production-readiness checklist

#	Production-readiness requirement	Status / evidence
1	Use case, owner, risk tier, benefit metrics and prohibited uses approved.	Required before go-live
2	Architecture, threat model, privacy and responsible-AI assessments complete.	Required before go-live
3	Source data/products/corpora/evaluation sets authorized, classified, regionalized and lineage-complete.	Required before go-live
4	Project, APIs, service agents, identities, IAM and separation of duties reviewed.	Required before go-live
5	Network paths, egress, private access, VPC Service Controls and dependent services tested.	Required before go-live
6	Encryption/CMEK and key rotation/deletion protection validated where applicable.	Required before go-live
7	Approved model alias and fallback model/location documented and evaluated.	Required before go-live
8	Model/prompt/corpus/index/agent/tool/runtime release manifest immutable.	Required before go-live
9	Offline quality, grounding, citation, safety, bias, security and cost tests pass.	Required before go-live
10	Performance, concurrency, quota, retry, timeout and circuit-breaker tests pass.	Required before go-live
11	RAG ACL filters, deletion propagation, source precedence and cache isolation pass.	Required before go-live
12	Agent tool schemas, server authorization, idempotency, approval and emergency disable pass.	Required before go-live

#	Production-readiness requirement	Status / evidence
13	Model/prompt/agent cards and independent validation evidence complete as required.	Required before go-live
14	SLOs, dashboards, alerts, on-call, runbooks and incident playbooks active.	Required before go-live
15	Prompt/evidence/tool telemetry redaction, access and retention approved.	Required before go-live
16	Budgets, anomaly alerts and unit-cost dashboard active.	Required before go-live
17	Model/prompt/index/agent rollback and regional/degraded fallback exercised.	Required before go-live
18	Corpus/index rebuild and source restoration exercise completed for Tier 0/1.	Required before go-live
19	User disclosure, human review, escalation, appeal/correction and accessibility implemented.	Required before go-live
20	Support, quota escalation, vendor/model retirement and periodic review process documented.	Required before go-live

32.3 Cross-volume dependencies

Volume	Dependency consumed by Volume 07	Volume 07 output used later
01 Executive	Business principles, NFRs, target architecture and global decisions.	AI-specific decisions and portfolio roadmap.
02 Landing Zone	Organization/projects, Shared VPC, identity, KMS, logging and VPC SC.	AI project/service/network/IAM requirements for expansion.
03 Data Platform	Regional lakehouse, BigQuery, metadata, classification, quality and sharing.	Training/retrieval/feature/evaluation consumption patterns.
04 Data Engineering	Pipeline standards, retries, replay, testing and orchestration.	AI/ML pipeline components and document/embedding workflows.
05 Data Mesh	Domain ownership, product contracts, SLOs and marketplace.	Domain AI product ownership and knowledge contracts.
06 Analytics	Certified KPIs, Looker/BigQuery semantic layer and personas.	Agent/AI calls to certified analytics interfaces.
08 Security	Detailed Zero Trust controls and regulatory mapping.	AI threat/control requirements and attack scenarios.
09 DevSecOps	CI/CD, supply chain, promotion and rollback.	AI release manifests, evaluation gates and pipeline examples.
10 FinOps	Cost allocation, budgets and optimization.	AI unit economics and cost drivers.
11 SRE	SLIs/SLOs, alerting, incident and readiness.	AI-specific signals, runbooks and error budgets.
12 DR	Tiered recovery and exercise governance.	Model/corpus/index/agent recovery patterns.
13 Terraform	Golden modules, state and policy-as-code.	AI foundation/resource module requirements.

32.4 Open architecture decisions and next actions

Type	Item	Owner / due
Open decision	Approve initial model catalog, aliases, model-routing policy and external model exception process.	AI Governance / Wave 0.
Open decision	Confirm regional availability/control matrix for Agent Runtime, Agent Search, RAG Engine and selected vector backends.	Architecture/Security / before design approval.
Open decision	Select first RAG backend and source connector pattern for each pilot region.	AI Platform/Knowledge owners / Wave 3.
Open decision	Set risk-tier metric thresholds and independent validation scope.	Responsible AI/Risk / Wave 0-2.
Open decision	Define AI trace content, redaction and retention by classification.	Security/Privacy/Records / Wave 1.
Open decision	Select durable state/memory store and DR tier for agent workflows.	Application Architecture/SRE / before agent pilot.
Next action	Approve ADR-AI-001 through ADR-AI-010.	Enterprise Architecture Review Board.
Next action	Choose one predictive ML, one employee RAG and one bounded read-only agent pilot.	CDO/CIO and domain owners.
Next action	Run current Google Cloud product, region, quota, security and provider validation.	AI Platform Architecture.
Next action	Build the nonproduction foundation and execute security/evaluation/rollback proofs before production commitment.	Platform, Security, SRE and pilot teams.

Appendix A - Artifact Inventory

Artifact	Location / purpose
Detailed guide	documents/Volume_07_AI_GenAI_RAG_LLMOps_Detailed.docx and .pdf
Architecture diagrams	diagrams/ in PNG, SVG, Mermaid, PlantUML, Graphviz DOT and draw.io formats
RAG example	examples/rag/
Vertex AI Pipeline example	examples/pipelines/
Prompt and evaluation examples	examples/prompts/
ADK agent example	examples/agents/
Terraform foundation example	examples/terraform/
Model-card example	examples/governance/
Matrices	matrices/ use-case, risk, access, SLO, evaluation, evidence and RACI CSVs
Package controls	README.md, MANIFEST.txt and SHA256SUMS.txt

Appendix B - Diagram Construction Specification

Each draw.io file is editable. Replace generic nodes with current official Google Cloud product icons while retaining the logical architecture. Use regional swimlanes for AMER, EMEA and APAC; show environment boundaries, VPC Service Controls perimeters, trust boundaries, identities, data flow, control flow and human-approval points. Split complex views rather than producing one unreadable diagram.

Diagram	Required construction emphasis
Enterprise AI platform	Experience, agent, model, RAG, MLOps/LLMOps, data, trust and operations planes.
Enterprise RAG	Source/ACL lineage, ingestion/chunking, embedding/index, authorization filter, rerank, grounding/citations and guardrails.
LLMOps lifecycle	Risk intake, design, evaluation, red team, approval, release, online monitoring and feedback.
MLOps architecture	Features/data, experiments, pipelines, registry, validation, deployment, monitoring and rollback.
Agent architecture	Identity/policy gateway, orchestrator/subagents, tool gateway, systems, human approval and trace.
AI security	Identity, perimeter, data, model/prompt, input/retrieval, output/action, detection and assurance.
Governance workflow	Registration, classification, design review, build/test, independent validation, approval, operation and retirement.

Appendix C - Representative Implementation Examples

The package examples are production-oriented starters, not drop-in production solutions. Each example includes purpose, prerequisites, configuration, error handling, logging, security, testing, deployment, rollback and limitations. The RAG and agent SDK calls require current validation because Google Cloud has moved current agent/RAG documentation into Gemini Enterprise Agent Platform and SDK surfaces evolve.

Appendix D - Official Documentation Validation Set

- Gemini Enterprise Agent Platform overview - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.
- Agent Development Kit overview - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.
- Agent Runtime overview - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.
- Agent Search documentation - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.
- RAG Engine documentation - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.
- Vector Search documentation - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.

- Vertex AI Pipelines introduction - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.
- Model Registry introduction - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.
- Vertex AI Experiments introduction - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.
- ML Metadata introduction - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.
- Feature Store overview - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.
- Model Monitoring overview - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.
- Generative AI evaluation overview - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.
- Grounding overview - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.
- Safety and content filters - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.
- VPC Service Controls with Vertex AI - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.
- CMEK for Vertex AI - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.
- Document AI documentation - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.
- Speech-to-Text documentation - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.
- Text-to-Speech documentation - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.
- Cloud Vision documentation - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.
- Sensitive Data Protection documentation - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.
- BigQuery ML introduction - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.
- Terraform Google provider - validate current features, models, locations, quotas, IAM, networking, VPC Service Controls, CMEK, SDK/provider and commercial terms before implementation.

Appendix E - Terminology and Naming

Term	Definition / naming standard
AI product	A governed business capability using models, prompts, data/knowledge, software, controls, SLOs and ownership.
Model alias	Stable enterprise name resolving to an approved model/version/configuration.
Prompt package	Versioned system/developer instructions, schemas, model alias, generation config, guardrails and evaluation suite.
Corpus/index release	Source snapshot plus chunking, embedding, metadata/ACL and backend configuration.
Agent release	Agent/subagent code/config, memory policy, model alias, tool catalog, policy and evaluation evidence.
Project	acme-ai-<domain|platform|eval>-<env>-<region>-01.
Release ID	rel-<usecase>-<yyyyMMdd>-<gitsha>; recorded across traces, registry and deployment.
Classification	Public, Internal, Confidential, Restricted or Regulated inherited from series foundation.

Document conclusion. Volume 07 establishes the enterprise AI, GenAI, RAG, agent, MLOps and LLMOps architecture without regenerating or changing Volumes 01-06. Approval authorizes controlled pilots and implementation planning subject to the open decisions, risk-tier approvals and current Google Cloud capability validation.